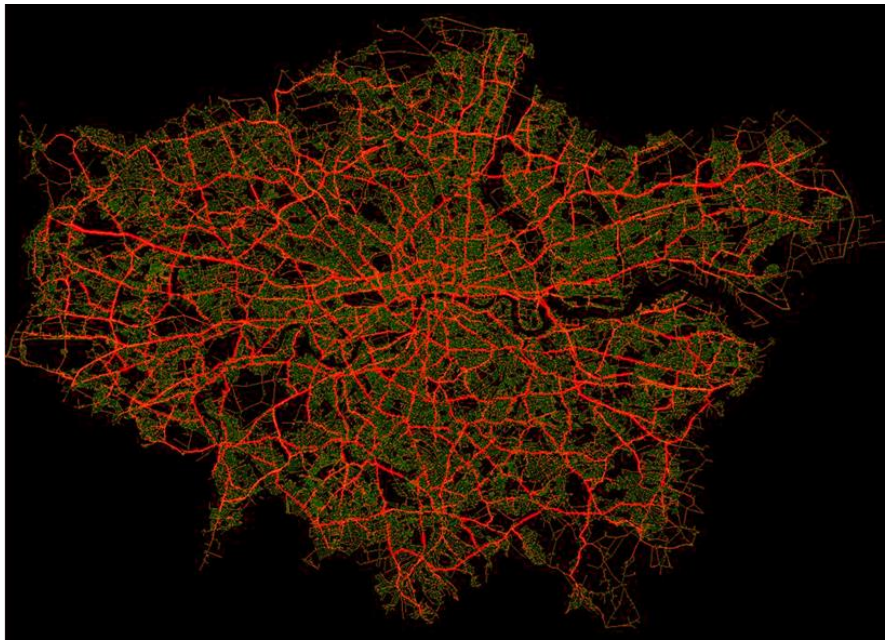
Transport and Mobility Modelling: Transport Demand (Part 1 of 2)

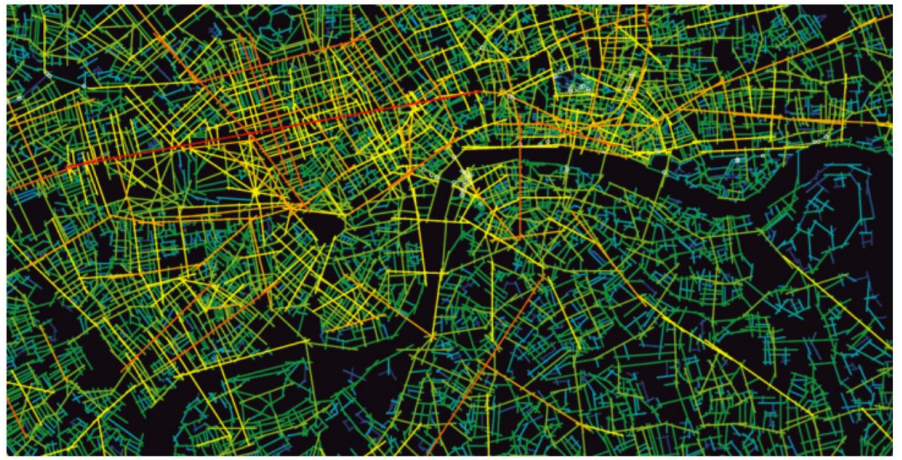
Anders Johansson

1. INTRO TO DEMAND MODELLING

Transport demand

- The demand for transport is a *derived* demand, i.e. it arises due to other needs (unlike demands for sports events, restaurants, etc).
- Transport is purposeful, i.e. people use transport systems in order to undertake certain activities in certain locations.
- Poor transport systems restrict access to opportunities, which limits economic and socio-economic development.





Top: Density of London road network.
Bottom: Space Syntax Axial Map

Demand modelling: The Four Step Model

Prerequisites:
- Tessellation / discretisation of space (e.g. postcodes, wards, boroughs, countries)
- Population / demographic data

Step 1: Trip generation: How many trips originate *O* (e.g. residential areas) and destinate *D* (e.g. industrial/business areas) in each zone. This can be estimated based on zonal attributes (e.g. population, demographic, land use)

Step 2: Trip distribution: Estimation of origin-destination pairings.

Step 3: Mode choice / modal split (walking, cycling, public transport, driving, rail, ...)

Step 4: Traffic assignment: Route choice.

Trip generation

The problem is to be able to estimate:

- O_i : The number of trips ‘generated’ in zone *i*.
- D_j : The number of trips ‘attracted to’ zone *j*.

This can be achieved in a number of ways, e.g.:

- Using descriptions of zones: Population size, car ownership ratio, employment, age distribution, or
- As a *trip frequency problem*: The probability that a person of a certain type will undertake *N* number of trips in a week.

Journey type

Normally, a different model for each type of journey is used, rather than mixing all types of trips together, e.g.:

- Travel to work
- Travel to school
- Shopping trips
- Leisure trips
- …

And also, different models for peak and off-peak are often used.

	AM Peak		Off Peak	
Purpose	**No.**	**%**	**No.**	**%**
Work	465 683	52.12	39 787	12.68
Education	313 275	35.06	15 567	4.96
Shopping	13 738	1.54	35 611	11.35
Social	7 064	0.79	16 938	5.40
Health	14 354	1.60	8 596	2.74
Bureaucracy	34 735	3.89	57 592	18.35
Accompanying	18 702	2.09	6 716	2.14
Other	1 736	0.19	2 262	0.73
Return to home	24 392	2.72	130 689	41.65

Modelling Transport – Juan de Dios Ortúzar, Luis G. Willumsen

In this lecture we will focus on the travel-to-work journey.

Trip **production** factors

The likelihood of a person making a certain trip is strongly dependent on their socioeconomic attributes, such as:

- Income
- Car ownership
- Family / household size
- Land value
- Residential density

Trip **attraction** factors

The likelihood of a person making a certain trip is strongly dependent on their socioeconomic attributes, such as:

- Employment level in destination zone
- Accessibility
- Total amount of floor space for industrial or commercial use

2. EMPIRICAL DATA: UK CENSUS

UK Census data

- “Gold standard” data
- Once every 10 years. Last one in 2011.
- “100%” sample.
- Two types of data of relevance to transport demand:
  - Population data
  - Interaction / flow data
- Interaction data:
  - Migration (to UK and within UK)
  - Journey-to-work / commuting data
  - School journey data
  - Movement to/from second residence

https://census.ukdataservice.ac.uk/use-data/guides/flow-data

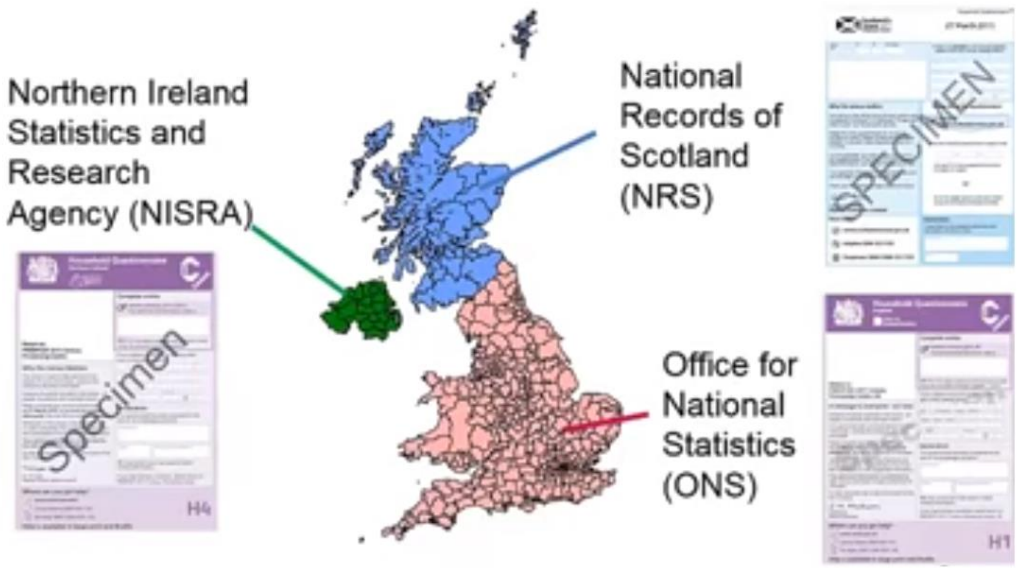
UK Census data

- Spatial units:
  - LA: Local Authority (e.g. ‘Bristol’)
  - Ward (35 in Bristol, e.g. ‘Clifton’)
  - MSOA: Middle Layer Super Output Area (53 in Bristol , 7,201 in England and Wales)
  - LSOA: Lower Layer Super Output Area (34,753 in England and Wales)
  - OA: Output area (181,408 in England and Wales)
  - Post codes (1.7 million in the UK)
- Which level is appropriate for your model depends on what questions you want to address (and how powerful your computer is)

ONS (Office for National Statistics):
https://www.ons.gov.uk/methodology/geography/ukgeographies/censusgeography

UK Census data: Complications and issues

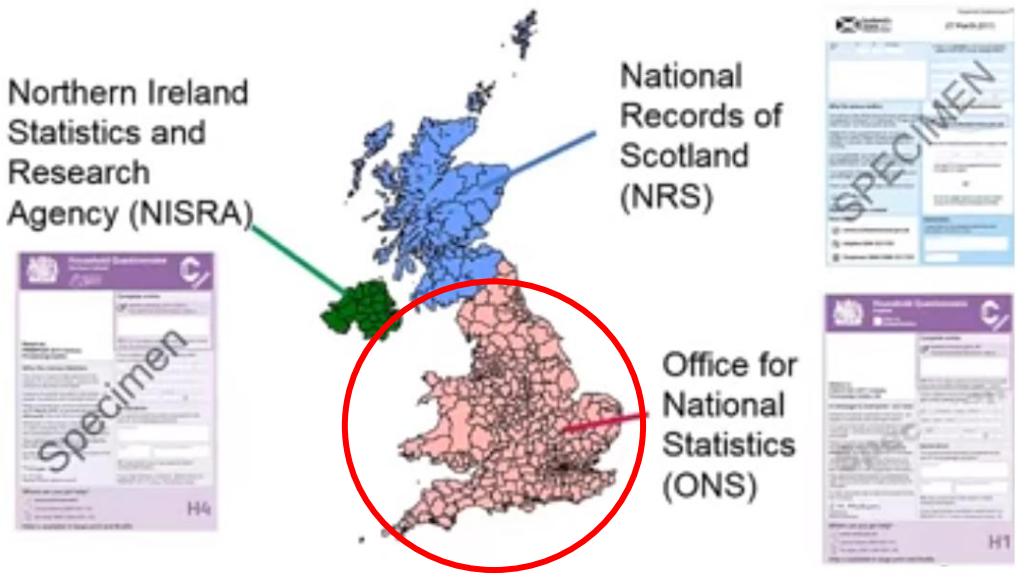
- Three different datasets combined (not identical questions)
- The definitions of zones and their boundaries change over time, and not necessarily in sync with the 10-year Census cycle
- There is a natural barrier (water) between GB and Northern Ireland
- Republic of Ireland not covered
- However, UK is an island(s) which makes it somewhat easier to manage cross-boundary flows
- A lot can happen in 10 years, and the Census only covers a 12-month period every 10 years.



https://census.ukdataservice.ac.uk/use-data/guides/flow-data

UK Census data: Complications and issues

- Three different datasets combined (not identical questions)
- The definitions of zones and their boundaries change over time, and not necessarily in sync with the 10-year Census cycle
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https://census.ukdataservice.ac.uk/use-data/guides/flow-data

Hands on: Empirical data

- Now, let us analyse some empirical data.
- *Data Science* in practice typically turns out to be 95% of the time being spent on gathering, transforming, filtering, and joining up different disparate data sets; and then 5% of the time actually doing the science.

Decide on what zonal definition to use

- First we need to decide what zonal definition to use.
  - Ward (35 in Bristol, e.g. ‘Clifton’)
  - **MSOA: Middle Layer Super Output Area (53 in Bristol , 7,201 in England and Wales)**
  - LSOA: Lower Layer Super Output Area (34,753 in England and Wales)
- MSOA is a good starting point as it is a manageable number of zones that will not put too much strain on your data analysis software (i.e. small enough that you can use Excel).

Obtain spatial / positional data

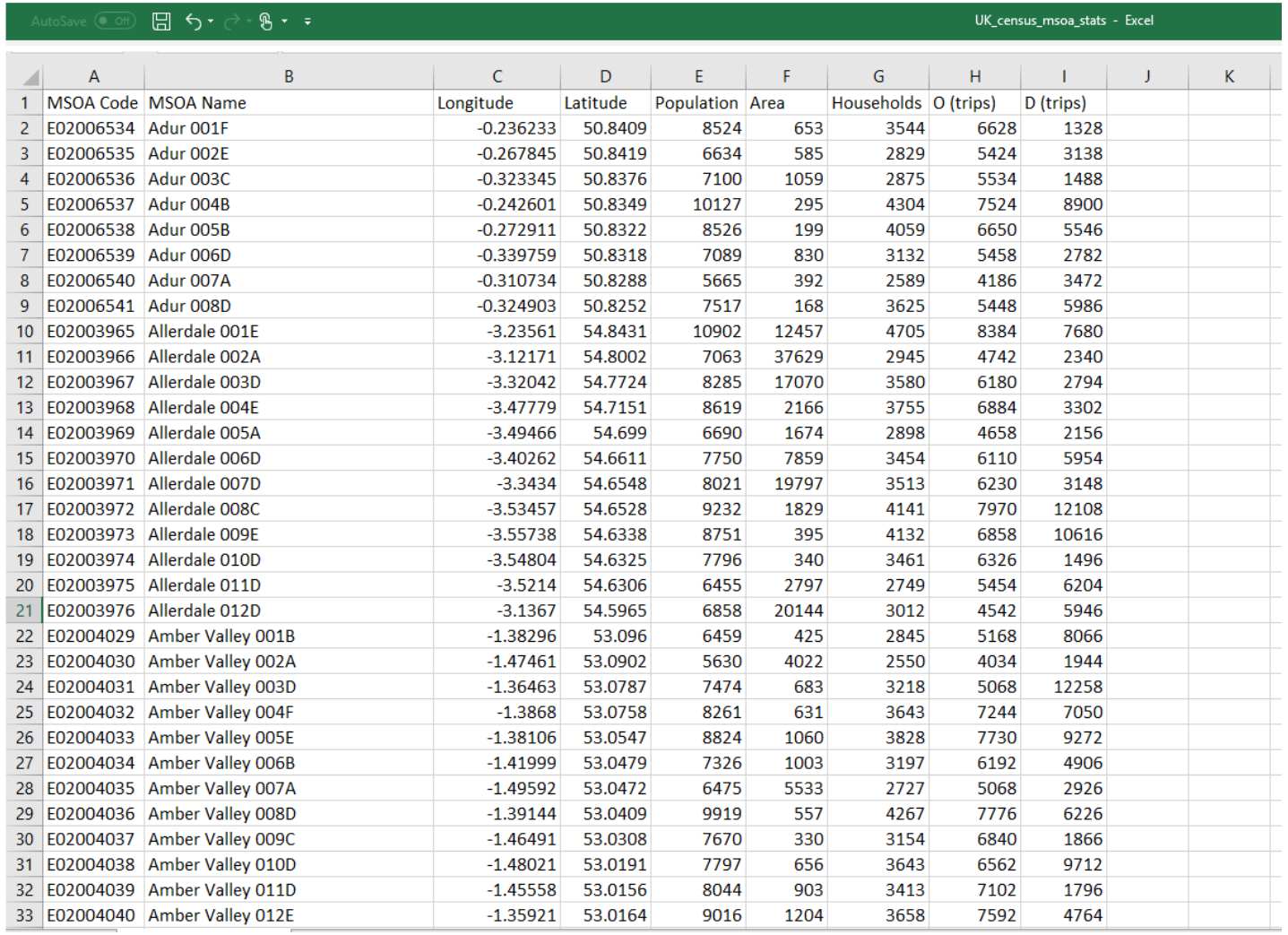
- Strictly speaking, spatial information (longitude/latitude) is not needed to estimate trips (O and D), but it is useful to have for context (and will be needed once we estimate O-D pairs later on.
- Unfortunately, there is no simple data set provided with lon/lat positions of all MSOA zones. Therefore, we need to be creative and
  - 1. Downlad a lon/lat data set for all postcodes in the UK
  - 2. Download a MSOA to postcodes mapping data set
  - 3. Use some data processing to calculate the average lon/lat of all postcodes that fall inside each of our MSOA zones, and then use that as the spatial reference.
- Alternatively we could have downloaded Government-provided GIS shapefiles of the boundaries of all MSOA zones, but then we would have needed GIS software and more involved computation.

Download reference O, D data for validation

- Luckily, the UK government openly share the Census data which includes the real reported O_i and D_j (travel to work journey), that we can use to test our models against.

UK Census O, D, and population data

- Putting everything together…

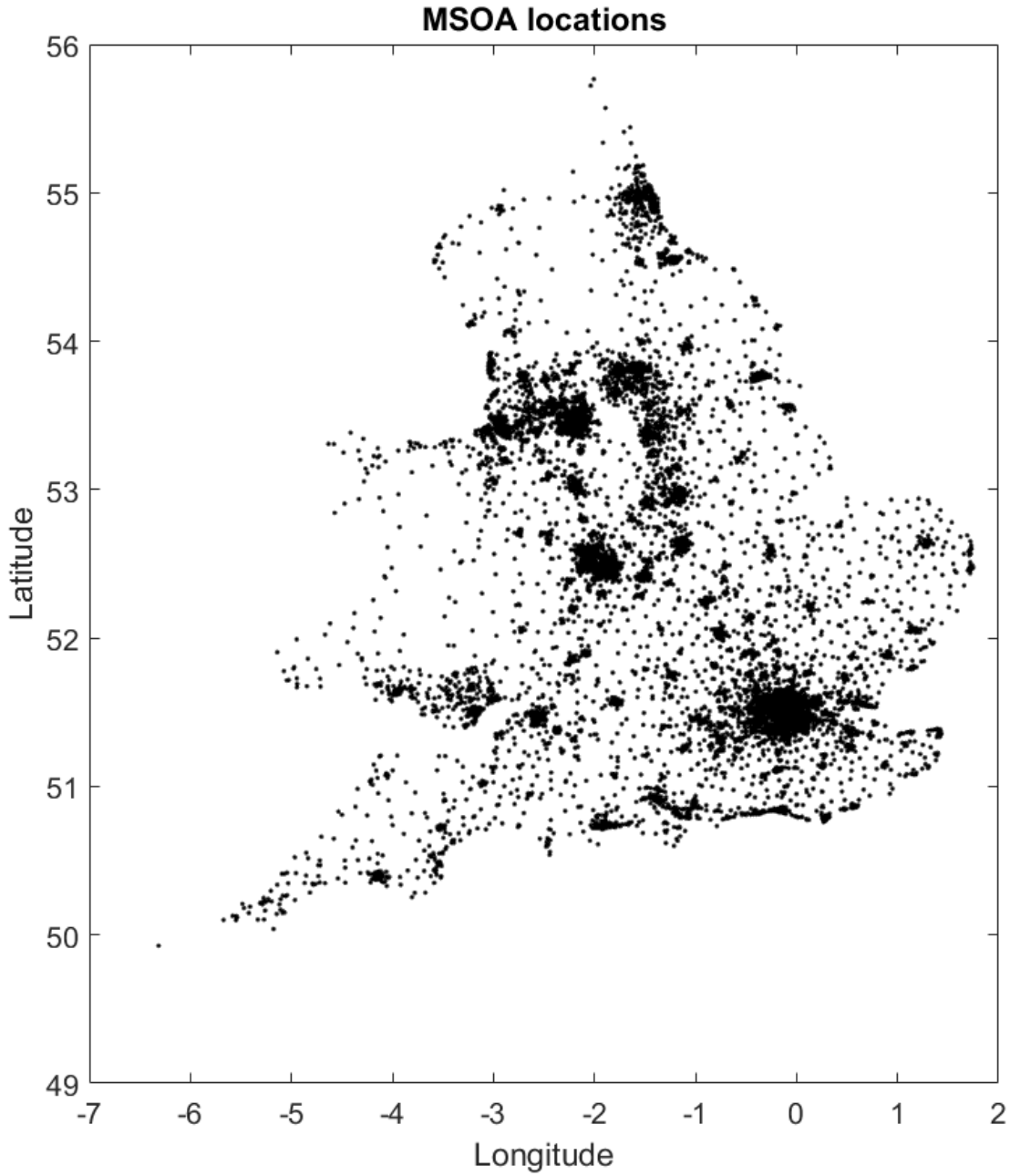


AutoSave Off — UK_census_msoa_stats - Excel

	A	B	C	D	E	F	G	H	I	J	K
1	MSOA Code	MSOA Name	Longitude	Latitude	Population	Area	Households	O (trips)	D (trips)		
2	E02006534	Adur 001F	-0.236233	50.8409	8524	653	3544	6628	1328		
3	E02006535	Adur 002E	-0.267845	50.8419	6634	585	2829	5424	3138		
4	E02006536	Adur 003C	-0.323345	50.8376	7100	1059	2875	5534	1488		
5	E02006537	Adur 004B	-0.242601	50.8349	10127	295	4304	7524	8900		
6	E02006538	Adur 005B	-0.272911	50.8322	8526	199	4059	6650	5546		
7	E02006539	Adur 006D	-0.339759	50.8318	7089	830	3132	5458	2782		
8	E02006540	Adur 007A	-0.310734	50.8288	5665	392	2589	4186	3472		
9	E02006541	Adur 008D	-0.324903	50.8252	7517	168	3625	5448	5986		
10	E02003965	Allerdale 001E	-3.23561	54.8431	10902	12457	4705	8384	7680		
11	E02003966	Allerdale 002A	-3.12171	54.8002	7063	37629	2945	4742	2340		
12	E02003967	Allerdale 003D	-3.32042	54.7724	8285	17070	3580	6180	2794		
13	E02003968	Allerdale 004E	-3.47779	54.7151	8619	2166	3755	6884	3302		
14	E02003969	Allerdale 005A	-3.49466	54.699	6690	1674	2898	4658	2156		
15	E02003970	Allerdale 006D	-3.40262	54.6611	7750	7859	3454	6110	5954		
16	E02003971	Allerdale 007D	-3.3434	54.6548	8021	19797	3513	6230	3148		
17	E02003972	Allerdale 008C	-3.53457	54.6528	9232	1829	4141	7970	12108		
18	E02003973	Allerdale 009E	-3.55738	54.6338	8751	395	4132	6858	10616		
19	E02003974	Allerdale 010D	-3.54804	54.6325	7796	340	3461	6326	1496		
20	E02003975	Allerdale 011D	-3.5214	54.6306	6455	2797	2749	5454	6204		
21	E02003976	Allerdale 012D	-3.1367	54.5965	6858	20144	3012	4542	5946		
22	E02004029	Amber Valley 001B	-1.38296	53.096	6459	425	2845	5168	8066		
23	E02004030	Amber Valley 002A	-1.47461	53.0902	5630	4022	2550	4034	1944		
24	E02004031	Amber Valley 003D	-1.36463	53.0787	7474	683	3218	5068	12258		
25	E02004032	Amber Valley 004F	-1.3868	53.0758	8261	631	3643	7244	7050		
26	E02004033	Amber Valley 005E	-1.38106	53.0547	8824	1060	3828	7730	9272		
27	E02004034	Amber Valley 006B	-1.41999	53.0479	7326	1003	3197	6192	4906		
28	E02004035	Amber Valley 007A	-1.49592	53.0472	6475	5533	2727	5068	2926		
29	E02004036	Amber Valley 008D	-1.39144	53.0409	9919	557	4267	7776	6226		
30	E02004037	Amber Valley 009C	-1.46491	53.0308	7670	330	3154	6840	1866		
31	E02004038	Amber Valley 010D	-1.48021	53.0191	7797	656	3643	6562	9712		
32	E02004039	Amber Valley 011D	-1.45558	53.0156	8044	903	3413	7102	1796		
33	E02004040	Amber Valley 012E	-1.35921	53.0164	9016	1204	3658	7592	4764		

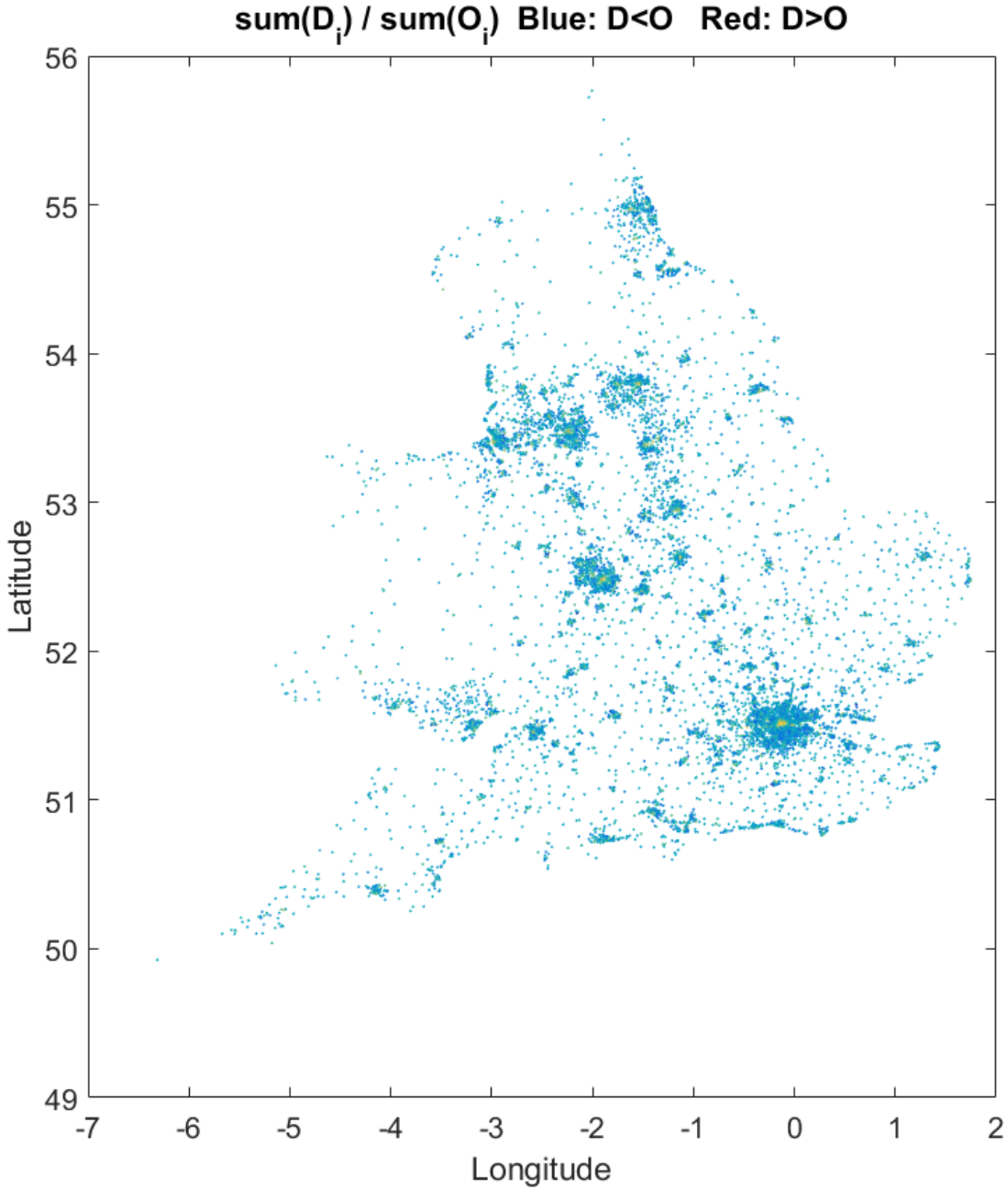
Longitude/latitude

- First of all, to validate that the data that we have obtained really do represent England and Wales, we plot the longitude and latitude values for all MSOA zones.



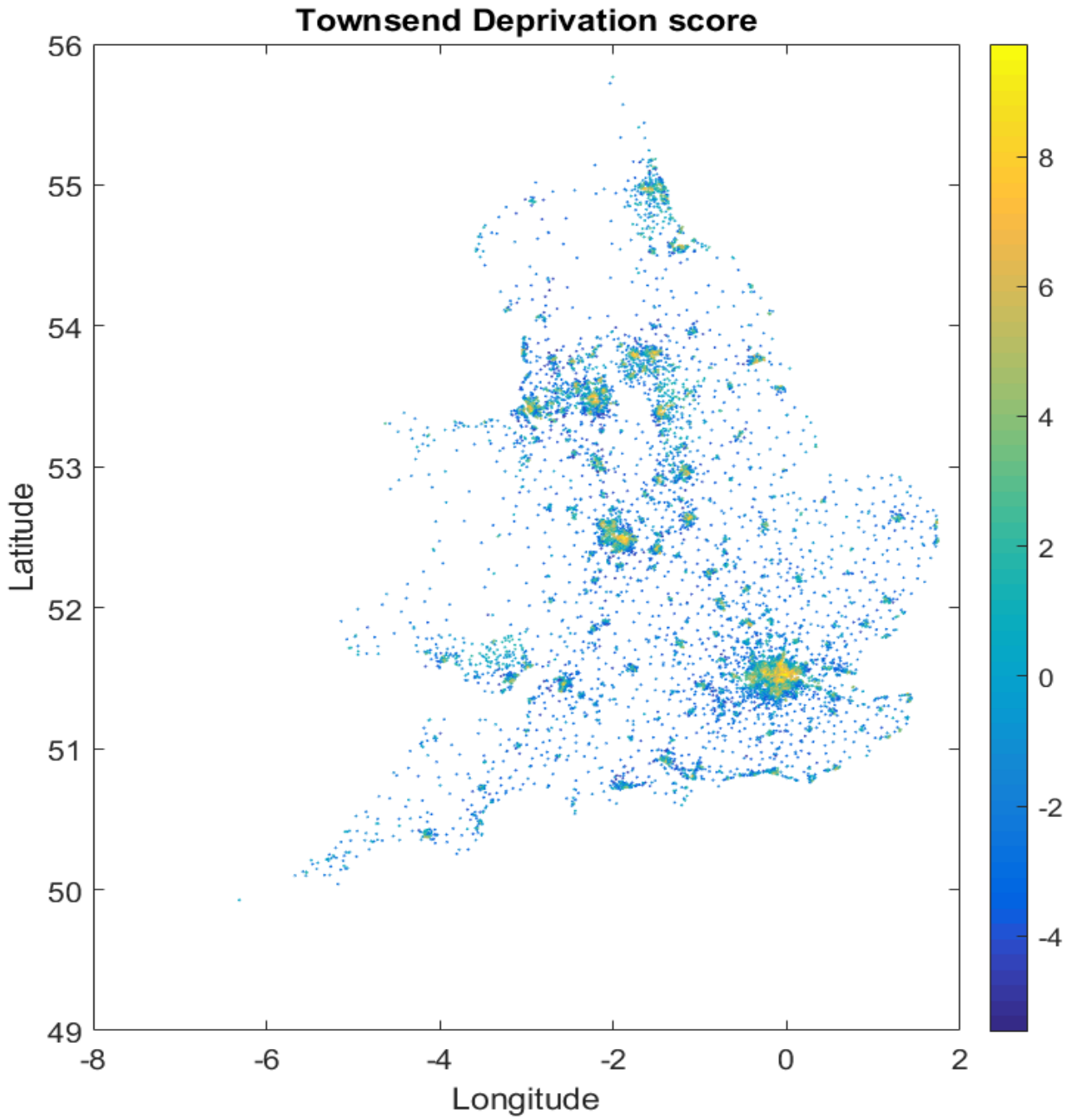
Attraction vs generation

- Next, we colour code each MSOA by D/O to see which zones are mainly trip generators and which are mainly trip attractors.



Deprivation

- The UK Census provides a range of useful socio-economic data sets, one of which is the Townsend Deprivation score.



3. ORIGIN / DESTINATION DATA

O and D histograms

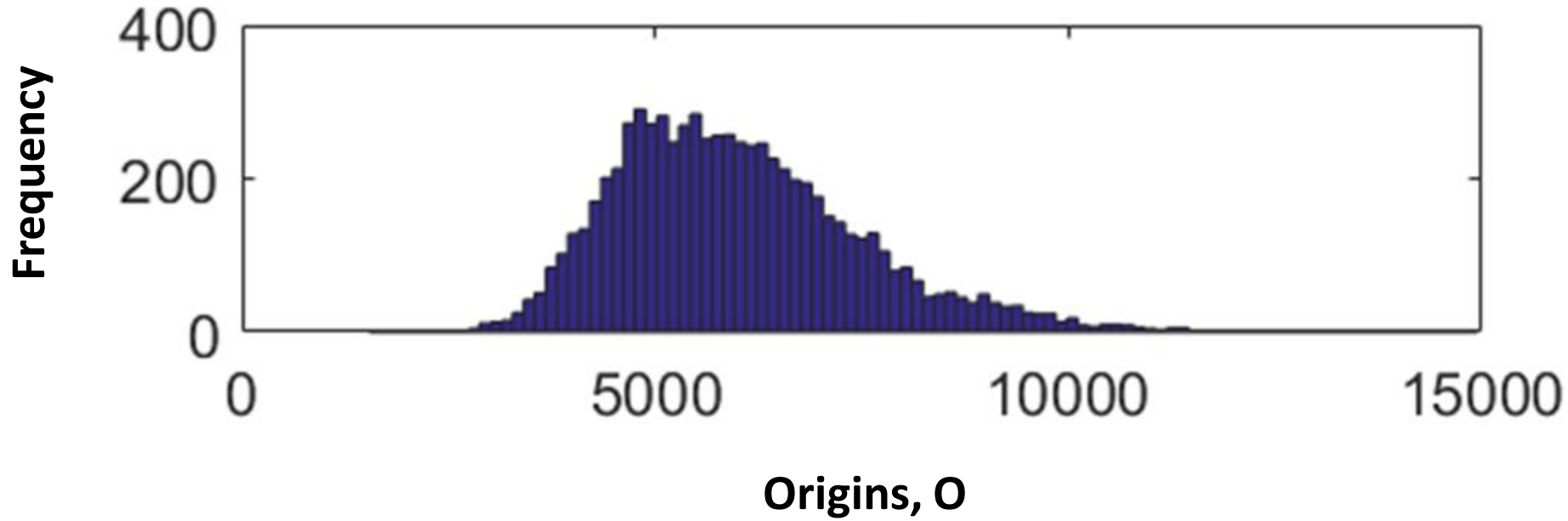
Now, let us get back to O and D:

- O_i : The number of trips ‘generated’ in zone *i*.
- D_j : The number of trips ‘attracted to’ zone *j*.

- **If we plot a histogram of O_i for all MSOA zones *i*, how do you expect it to look like?**

O and D histograms

- **If we plot a histogram of O_i for all MSOA zones *i*, how do you expect it to look like?**

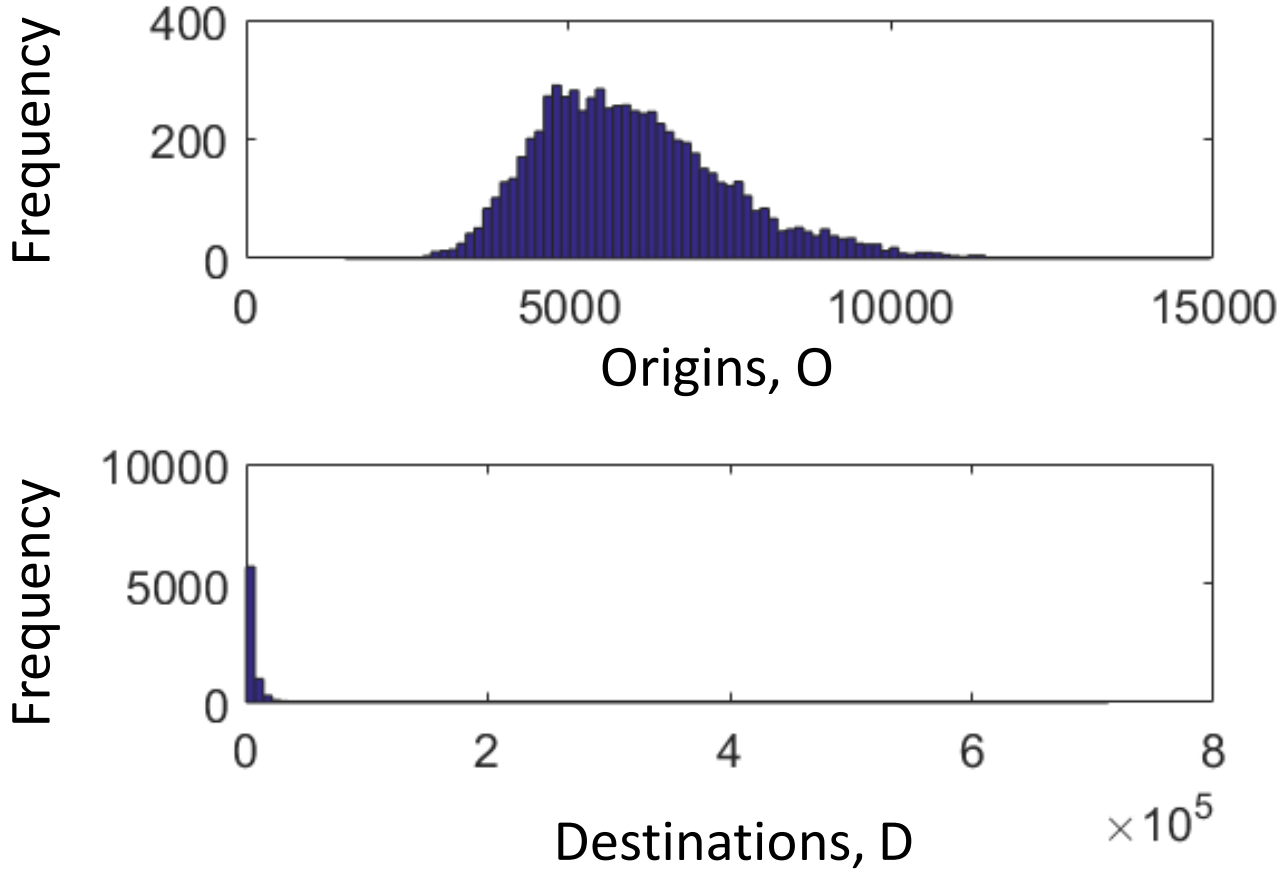


O and D histograms

- **Do you think a histogram of D will look essentially the same, or different in any way?**

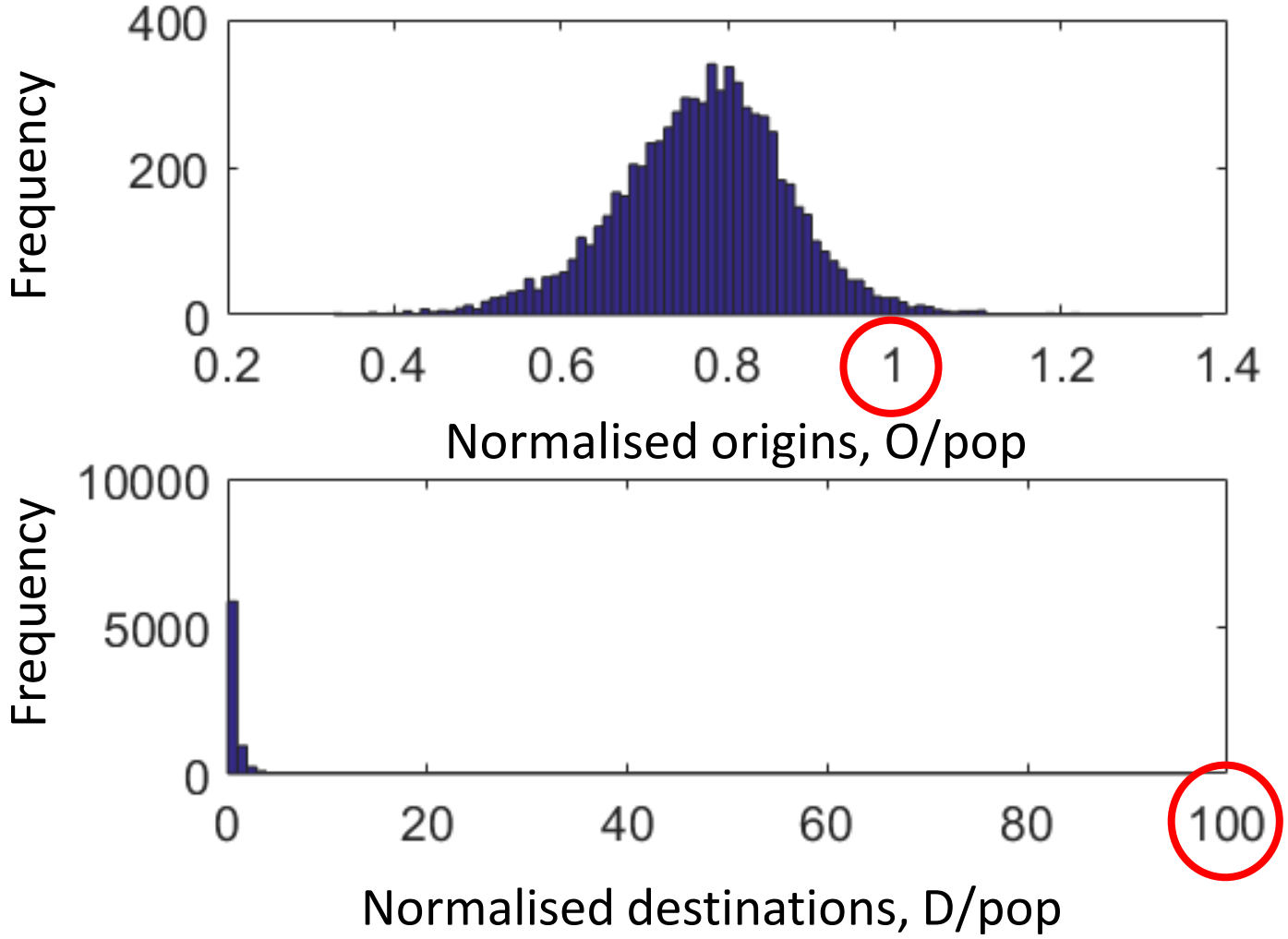
O and D histograms

- The histogram of O and the histogram of D for all MSOA zones show very different patterns.
- **Why is that?**



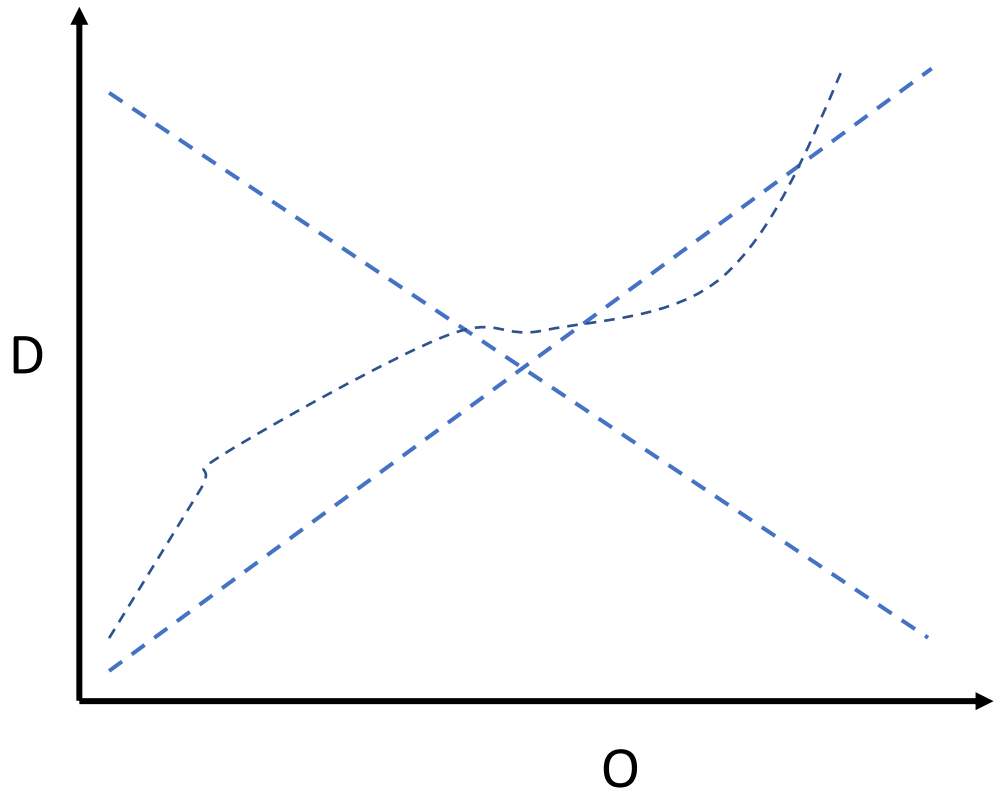
O and D histograms – normalised by pop.

- The histogram of O and the histogram of D for all MSOA zones show very different patterns.
- **What is the interpretation and significance of these two numbers?**



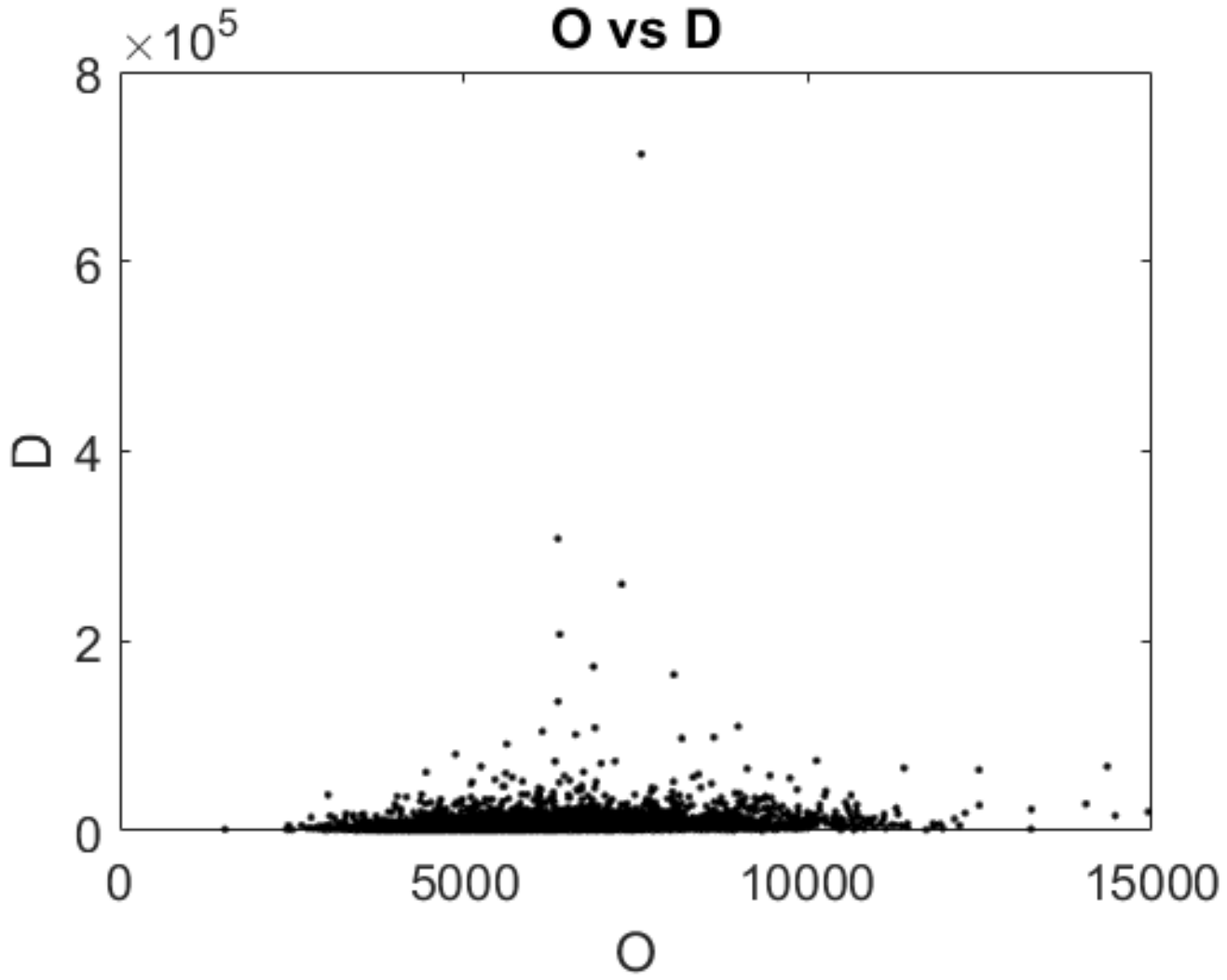
O vs D ?

- **What do you think an O vs D plot would look like? What – if any – shape might it have?**



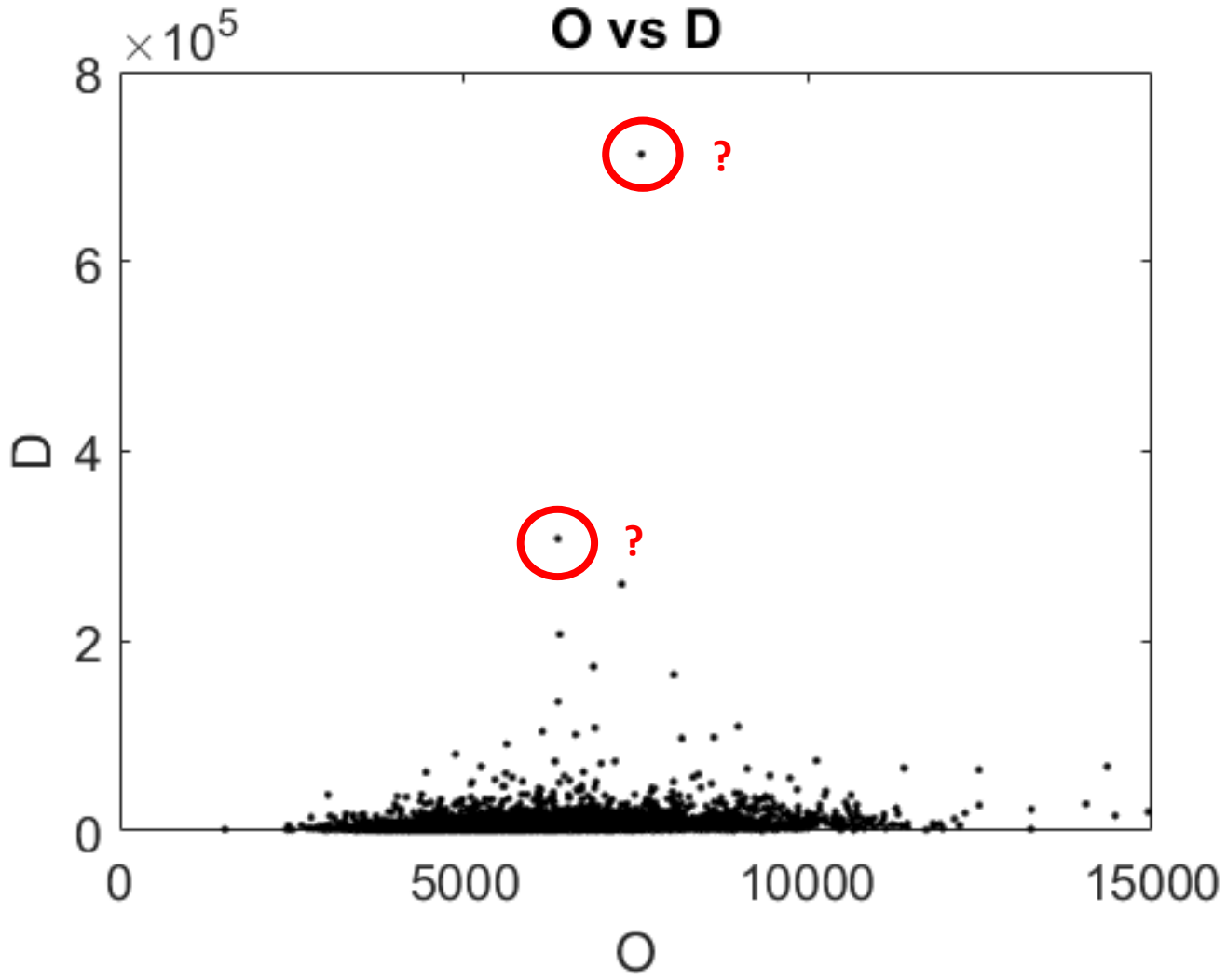
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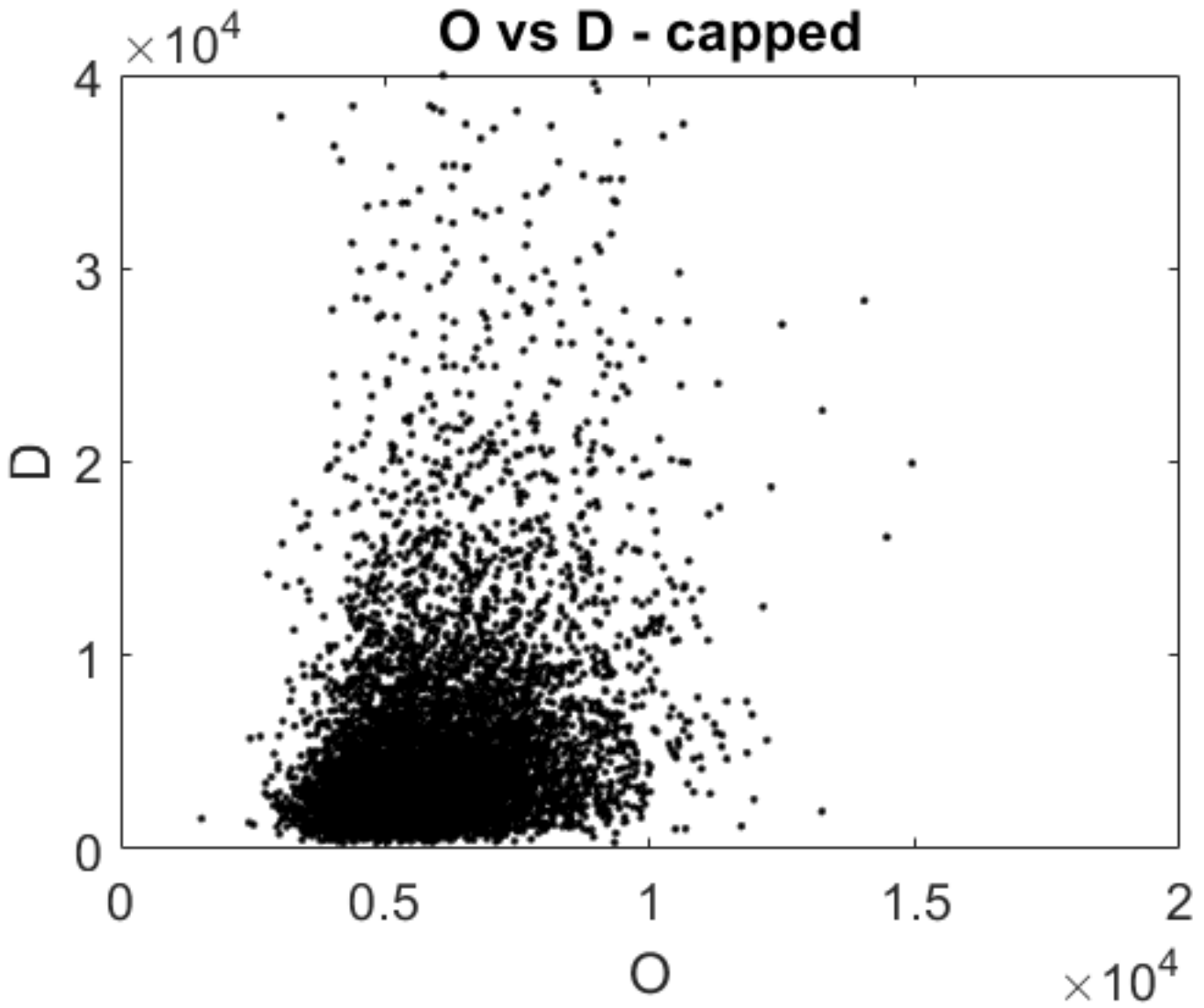
O vs D

- **What might these outliers be?**



O vs D

- Zoomed in / capped version
- **Why is there no clear trend?**

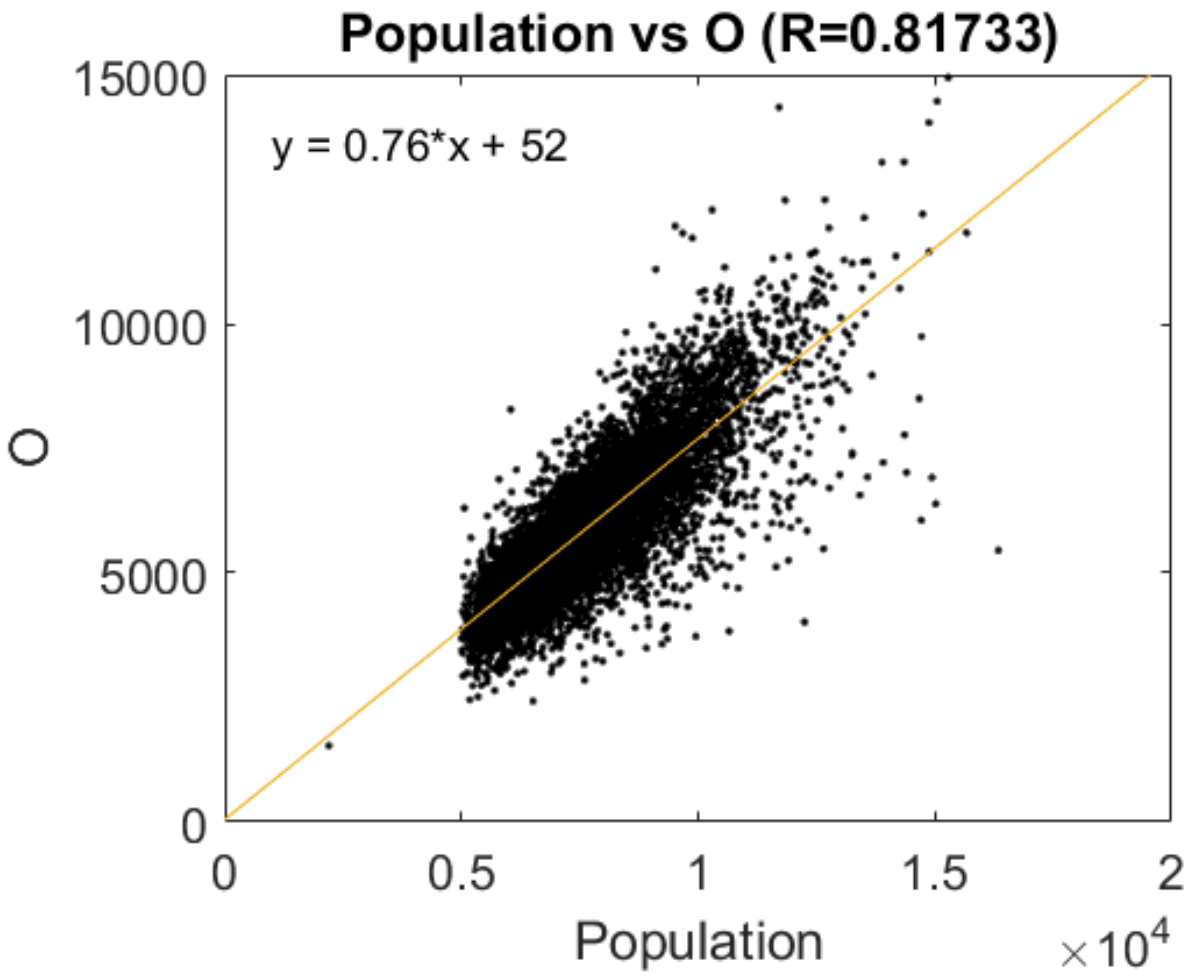


Models to predict O and D

- The UK Census has got very accurate estimates of *O* and *D* at very fine spatial scales.
- That is certainly not the case in most other parts of the world.
- Therefore, normally *O* and *D* need to be estimated using a model.
- Furthermore, by finding a model that can predict *O* and *D*, we will learn more about what the processes and mechanisms are that generate and attract trips.
- Let us start with trying to find a model to predict *O.*

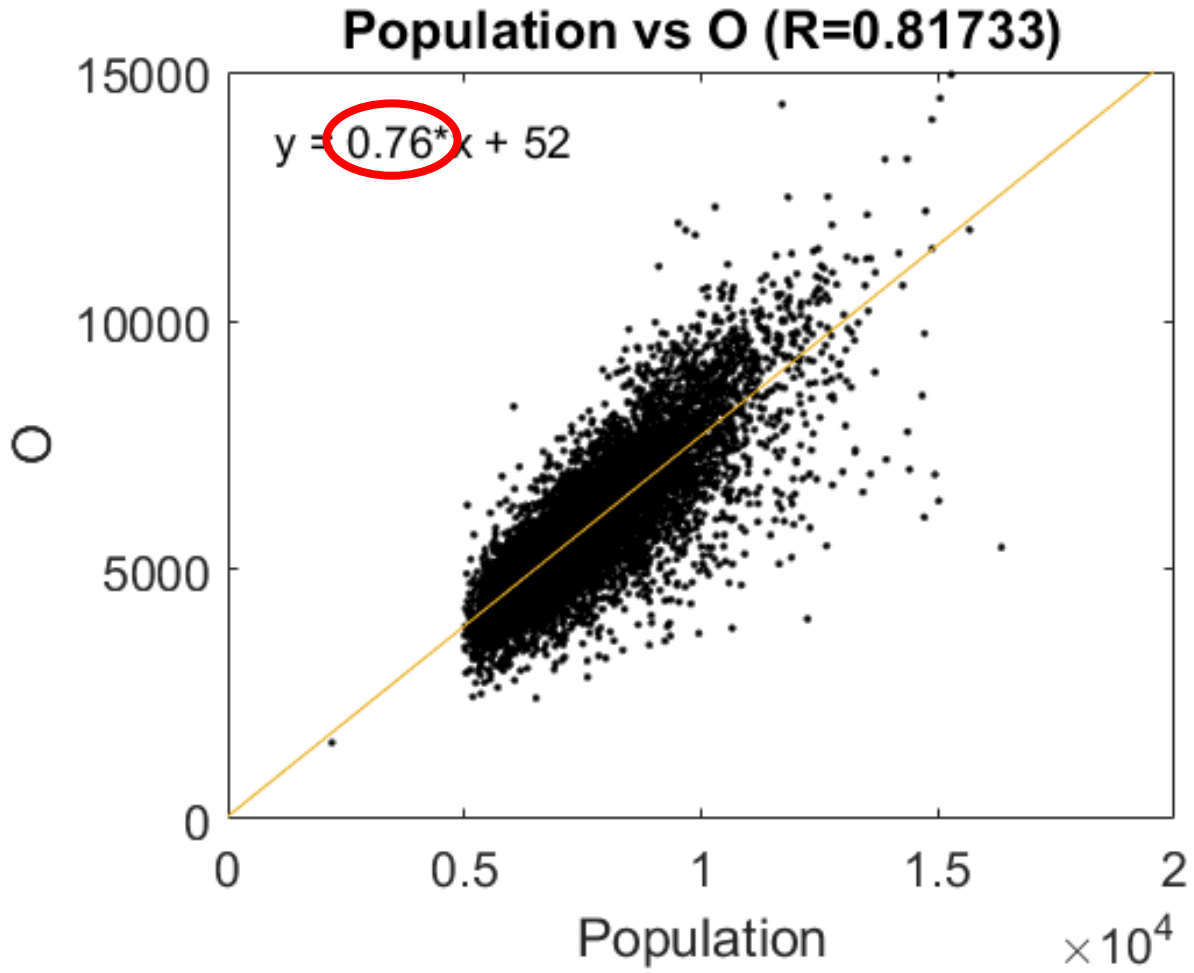
Population vs O

- As you can see, the zonal population alone provides a good estimate of how many trips a zone will generate.



Population vs O

- As you can see, the zonal population alone provides a good estimate of how many trips a zone will generate.
- **But what does the 0.76 slope represent?**



To summarise…

The number of origin trips O_i

- is proportional to the population in *i*.
- has got a natural upper limit which is the population in *i*.

The number of destination trips D_j

- has not got a natural upper limit.

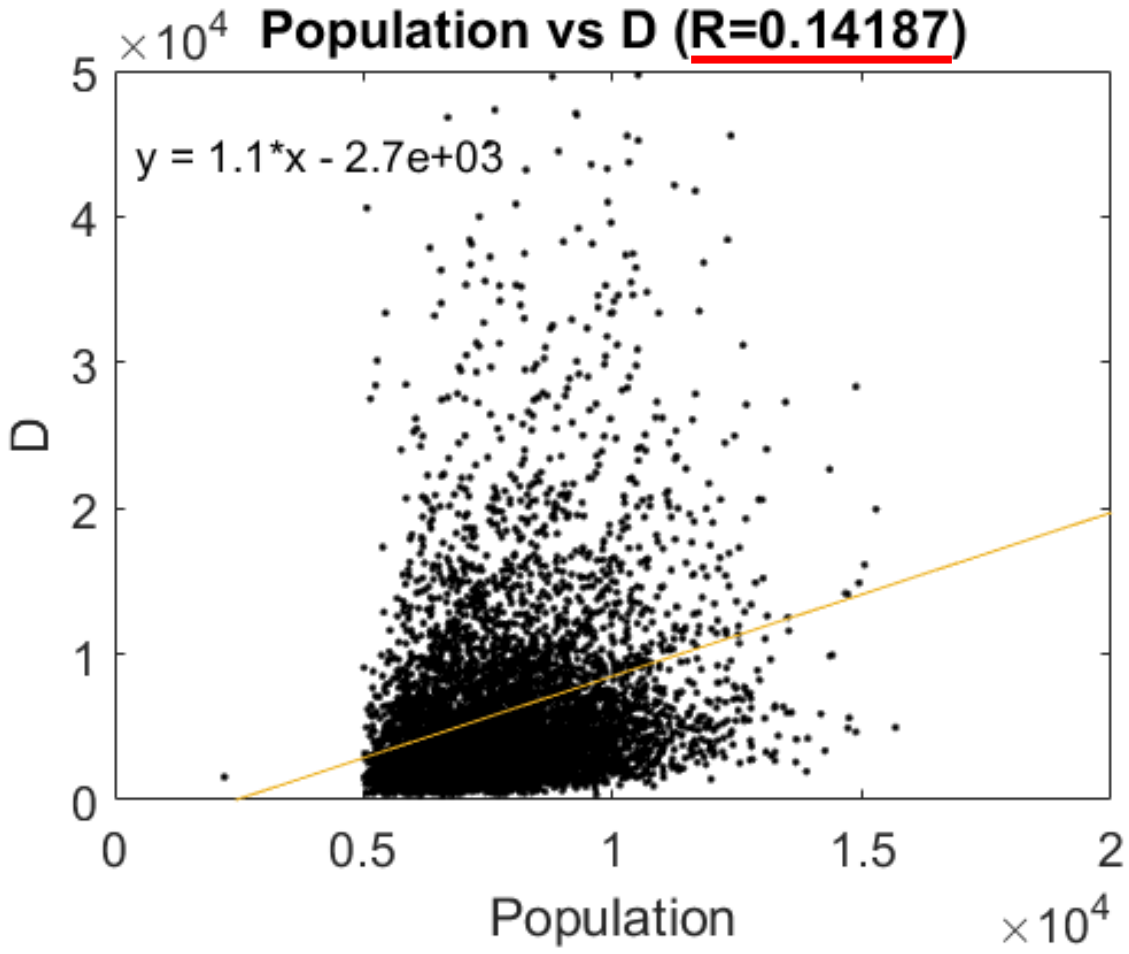
Tomorrow we will look at:

- models to predict *D.*
- the sensitivity of the results to the zonal size / definition.
- non travel-to-work journeys.
- other non-Census type data sets.
- temporal behaviour of demand.

4. Models to predict D_j

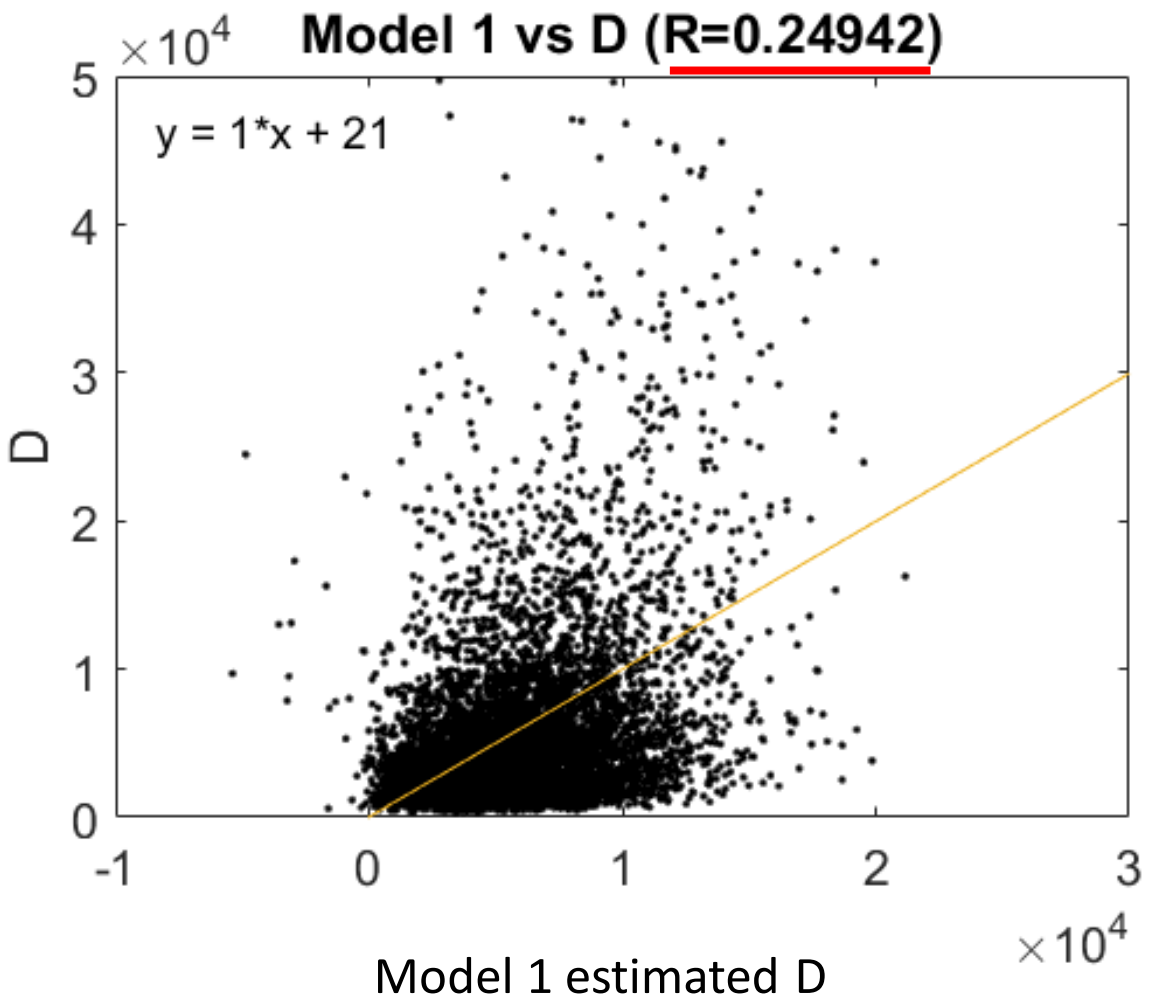
Population vs D

- MSOA population vs D
- **Why is this data set so much more noisy than the previous one?**
- What other data sources can we use to better predict destination trips D?



Model 1

- Model 1: System of linear equations *Ax=b*, based on a set of population, demographic and socioeconomic factors:
  - Longitude
  - latitude
  - Population
  - Area
  - Households
  - Deprivation

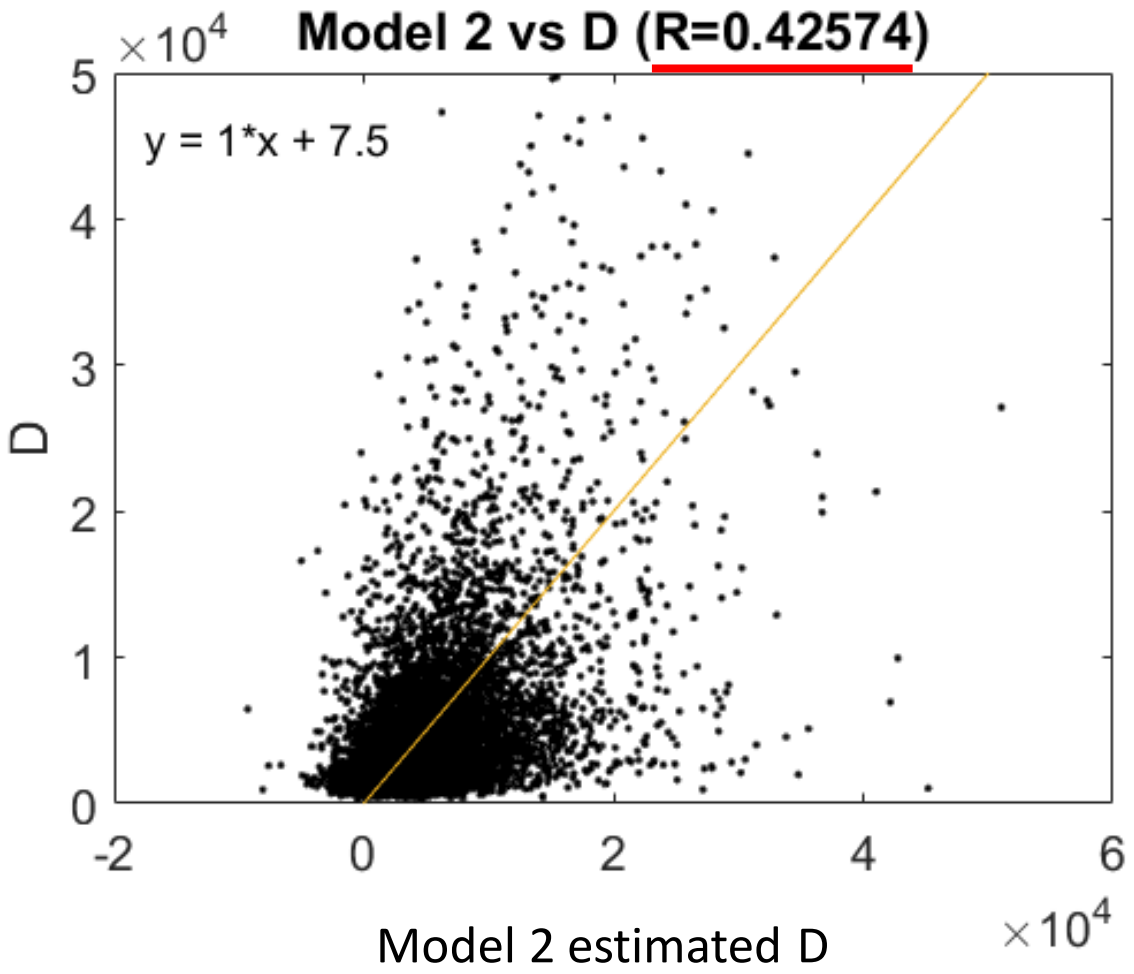


UK Census Employment categories

Occupation: All categories: Occupation; Industry: All categories: Industry; measures: Value
Occupation: All categories: Occupation; Industry: A, B, D, E Agriculture, energy and water; measures: Value",Occupation: All categories: Occupation; Industry: C Manufacturing; measures: Value,Occupation: All categories: Occupation; Industry: F Construction; measures: Value
Occupation: All categories: Occupation; Industry: G, I Distribution, hotels and restaurants; measures: Value"
Occupation: All categories: Occupation; Industry: H, J Transport and communication; measures: Value"
Occupation: All categories: Occupation; Industry: K, L, M, N Financial, Real Estate, Professional and Administrative activities; measures: Value"
Occupation: All categories: Occupation; Industry: O, P, Q Public administration, education and health; measures: Value"
Occupation: All categories: Occupation; Industry: R, S, T, U Other; measures: Value"
Occupation: 1. Managers, directors and senior officials; Industry: All categories: Industry; measures: Value"
Occupation: 1. Managers, directors and senior officials; Industry: A, B, D, E Agriculture, energy and water; measures: Value"
Occupation: 1. Managers, directors and senior officials; Industry: C Manufacturing; measures: Value"
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Occupation: 2. Professional occupations; Industry: R, S, T, U Other; measures: Value",Occupation: 3. Associate professional and technical occupations; Industry: All categories: Industry; measures: Value
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Occupation: 4. Administrative and secretarial occupations; Industry: K, L, M, N Financial, Real Estate, Professional and Administrative activities; measures: Value"
Occupation: 4. Administrative and secretarial occupations; Industry: O, P, Q Public administration, education and health; meas
Occupation: 4. Administrative and secretarial occupations; Industry: R, S, T, U Other; measures: Value",Occupation: 5. Skilled trades occupations; Industry: All categories: Industry; measures: Value
Occupation: 5. Skilled trades occupations; Industry: A, B, D, E Agriculture, energy and water; measures: Value",Occupation: 5. Skilled trades occupations; Industry: C Manufacturing; measures: Value,Occupation: 5. Skilled trades occupations; Industry: F Construction; measures: Value
Occupation: 5. Skilled trades occupations; Industry: G, I Distribution, hotels and restaurants; measures: Value"
Occupation: 5. Skilled trades occupations; Industry: H, J Transport and communication; measures: Value"
Occupation: 5. Skilled trades occupations; Industry: K, L, M, N Financial, Real Estate, Professional and Administrative activities; measures: Value"
Occupation: 5. Skilled trades occupations; Industry: O, P, Q Public administration, education and health; measures: Value"
Occupation: 5. Skilled trades occupations; Industry: R, S, T, U Other; measures: Value"
Occupation: 6. Caring, leisure and other service occupations; Industry: All categories: Industry; measures: Value"
Occupation: 6. Caring, leisure and other service occupations; Industry: A, B, D, E Agriculture, energy and water; measures: Value"
Occupation: 6. Caring, leisure and other service occupations; Industry: C Manufacturing; measures: Value"
Occupation: 6. Caring, leisure and other service occupations; Industry: F Construction; measures: Value"
Occupation: 6. Caring, leisure and other service occupations; Industry: G, I Distribution, hotels and restaurants; measures: Value"
Occupation: 6. Caring, leisure and other service occupations; Industry: H, J Transport and communication; measures: Value"
Occupation: 6. Caring, leisure and other service occupations; Industry: K, L, M, N Financial, Real Estate, Professional and Administrative activities; measures: Value"
Occupation: 6. Caring, leisure and other service occupations; Industry: O, P, Q Public administration, education and health; measures: Value"
Occupation: 6. Caring, leisure and other service occupations; Industry: R, S, T, U Other; measures: Value",Occupation: 7. Sales and customer service occupations; Industry: All categories: Industry; measures: Value
Occupation: 7. Sales and customer service occupations; Industry: A, B, D, E Agriculture, energy and water; measures: Value",Occupation: 7. Sales and customer service occupations; Industry: C Manufacturing; measures: Value,Occupation: 7. Sales and customer service occupations; Industry: F Construction; measures: Value
Occupation: 7. Sales and customer service occupations; Industry: G, I Distribution, hotels and restaurants; measures: Value"
Occupation: 7. Sales and customer service occupations; Industry: H, J Transport and communication; measures: Value"
Occupation: 7. Sales and customer service occupations; Industry: K, L, M, N Financial, Real Estate, Professional and Administrative activities; measures: Value"
Occupation: 7. Sales and customer service occupations; Industry: O, P, Q Public administration, education and health; measures: Value"
Occupation: 7. Sales and customer service occupations; Industry: R, S, T, U Other; measures: Value"
Occupation: 8. Process, plant and machine operatives; Industry: All categories: Industry; measures: Value"
Occupation: 8. Process, plant and machine operatives; Industry: A, B, D, E Agriculture, energy and water; measures: Value"
Occupation: 8. Process, plant and machine operatives; Industry: C Manufacturing; measures: Value"
Occupation: 8. Process, plant and machine operatives; Industry: F Construction; measures: Value"
Occupation: 8. Process, plant and machine operatives; Industry: G, I Distribution, hotels and restaurants; measures: Value"
Occupation: 8. Process, plant and machine operatives; Industry: H, J Transport and communication; measures: Value"
Occupation: 8. Process, plant and machine operatives; Industry: K, L, M, N Financial, Real Estate, Professional and Administrative activities; measures: Value"
Occupation: 8. Process, plant and machine operatives; Industry: O, P, Q Public administration, education and health; measures: Value"
Occupation: 8. Process, plant and machine operatives; Industry: R, S, T, U Other; measures: Value",Occupation: 9. Elementary occupations; Industry: All categories: Industry; measures: Value
Occupation: 9. Elementary occupations; Industry: A, B, D, E Agriculture, energy and water; measures: Value",Occupation: 9. Elementary occupations; Industry: C Manufacturing; measures: Value,Occupation: 9. Elementary occupations; Industry: F Construction; measures: Value
Occupation: 9. Elementary occupations; Industry: G, I Distribution, hotels and restaurants; measures: Value"
Occupation: 9. Elementary occupations; Industry: H, J Transport and communication; measures: Value"
Occupation: 9. Elementary occupations; Industry: K, L, M, N Financial, Real Estate, Professional and Administrative activities; measures: Value"
Occupation: 9. Elementary occupations; Industry: O, P, Q Public administration, education and health; measures: Value"
Occupation: 9. Elementary occupations; Industry: R, S, T, U Other; measures: Value"

Population vs D

- Model 2: Same as Model 1 but also include all ~100 employment categories.



What to do next…?

We could continue to heap data onto the problem until the R value is reasonably high, but:

- Is that a good thing to do?
- Is there another approach we could take?

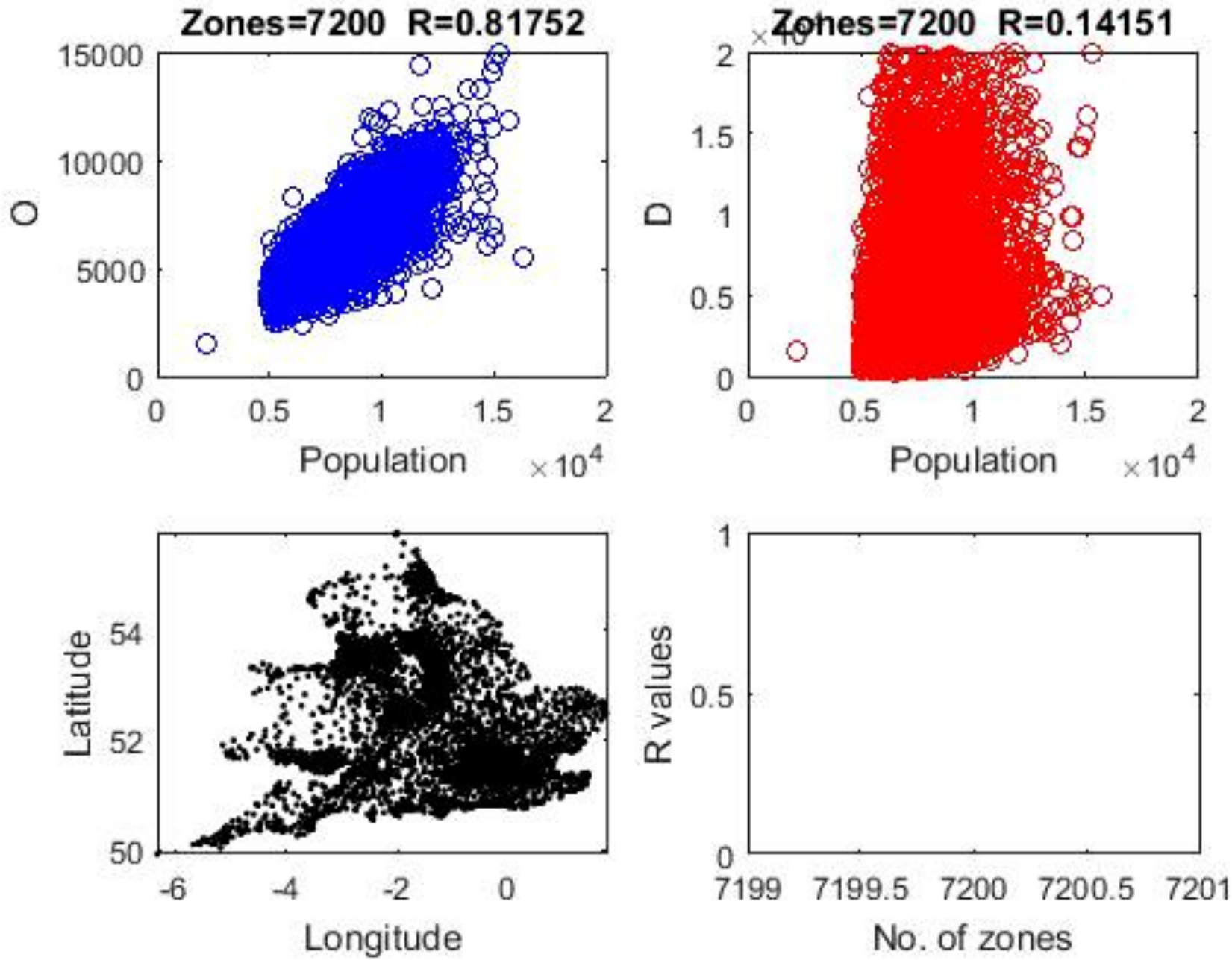
Sensitivity to zonal definition

To allow us to analyse the sensitivity of our results to the zonal definition, we will now run the following algorithm on the data:

- Pick a random zone *i*.
- Locate the nearest neighbour zone *j*.
- Merge zone *j* with zone *i*.
- Repeat until the number of zones has reached your target number.

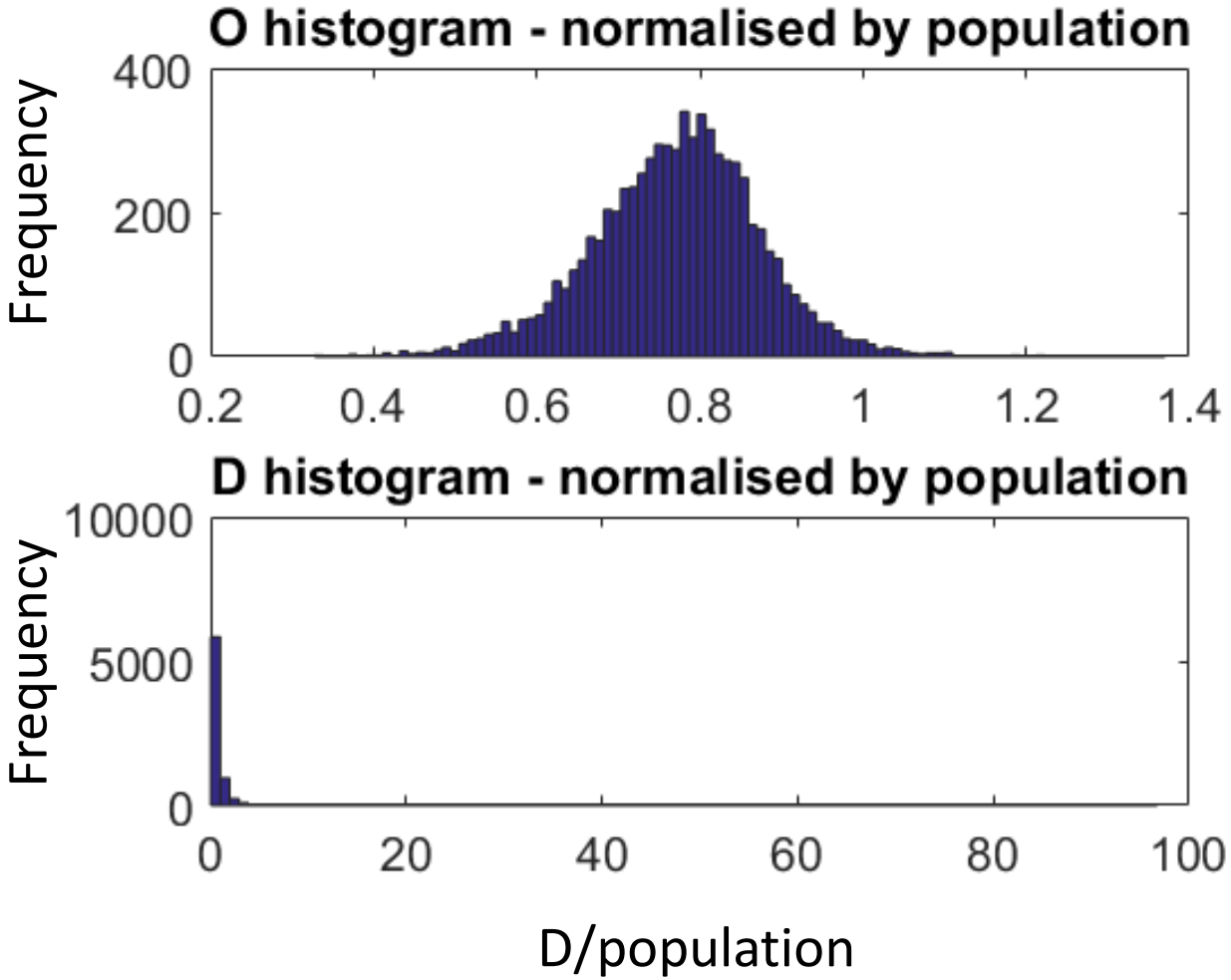
Sensitivity to zonal definition

If we now run our new zone-merging algorithm onto our Census / MSOA dataset, do you think that the results will look *more* or *less* noisy?


Zones=7200 R=0.81752
O
Population
×10^4
Zones=7200 R=0.14151
D
Population
×10^4
Latitude
Longitude
R values
No. of zones

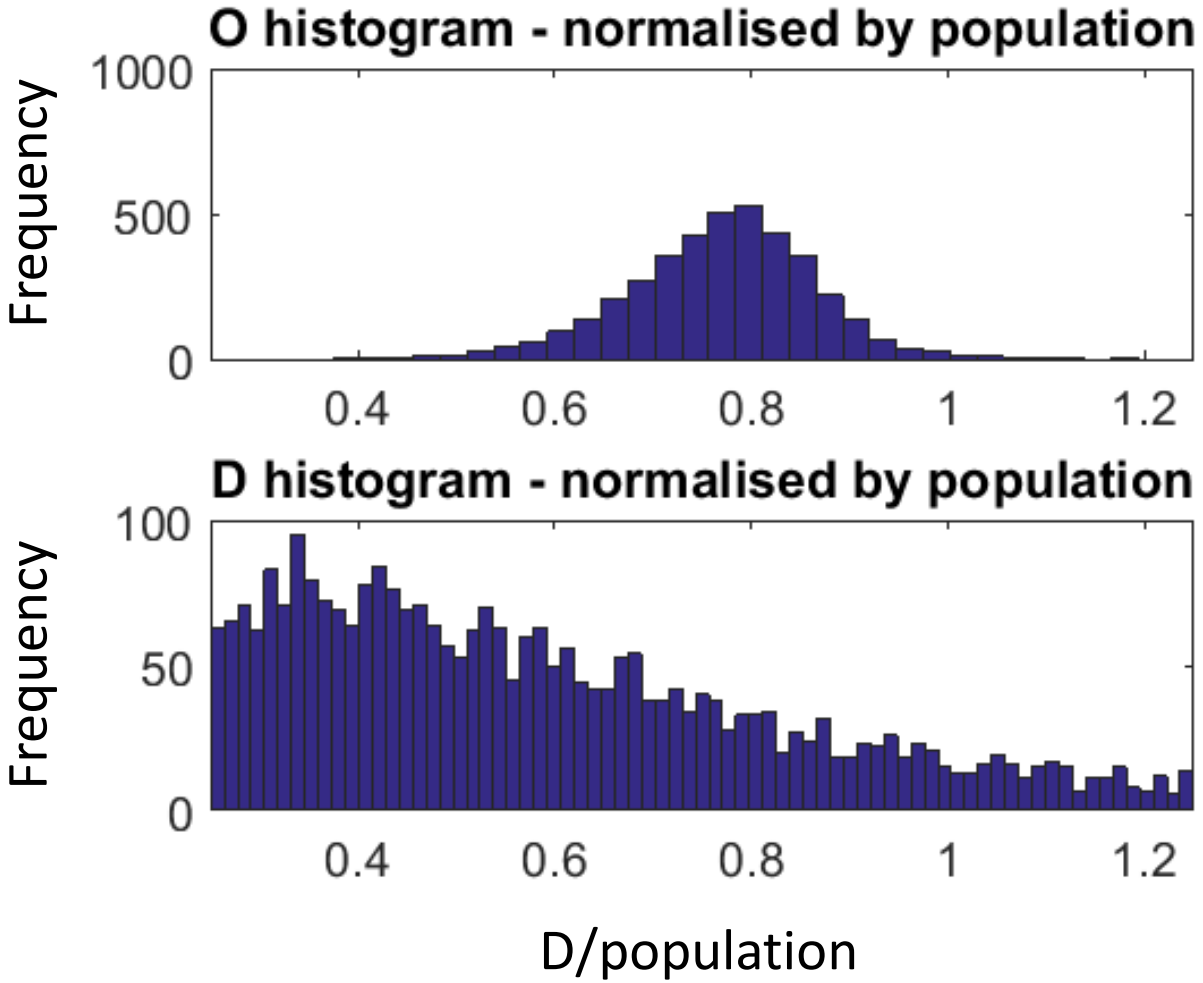
O and D histograms – normalised by pop.

- Let us now revisit the normalized O and D histograms with our merged zones.
- N = 7200 zones (original MSOA zones)



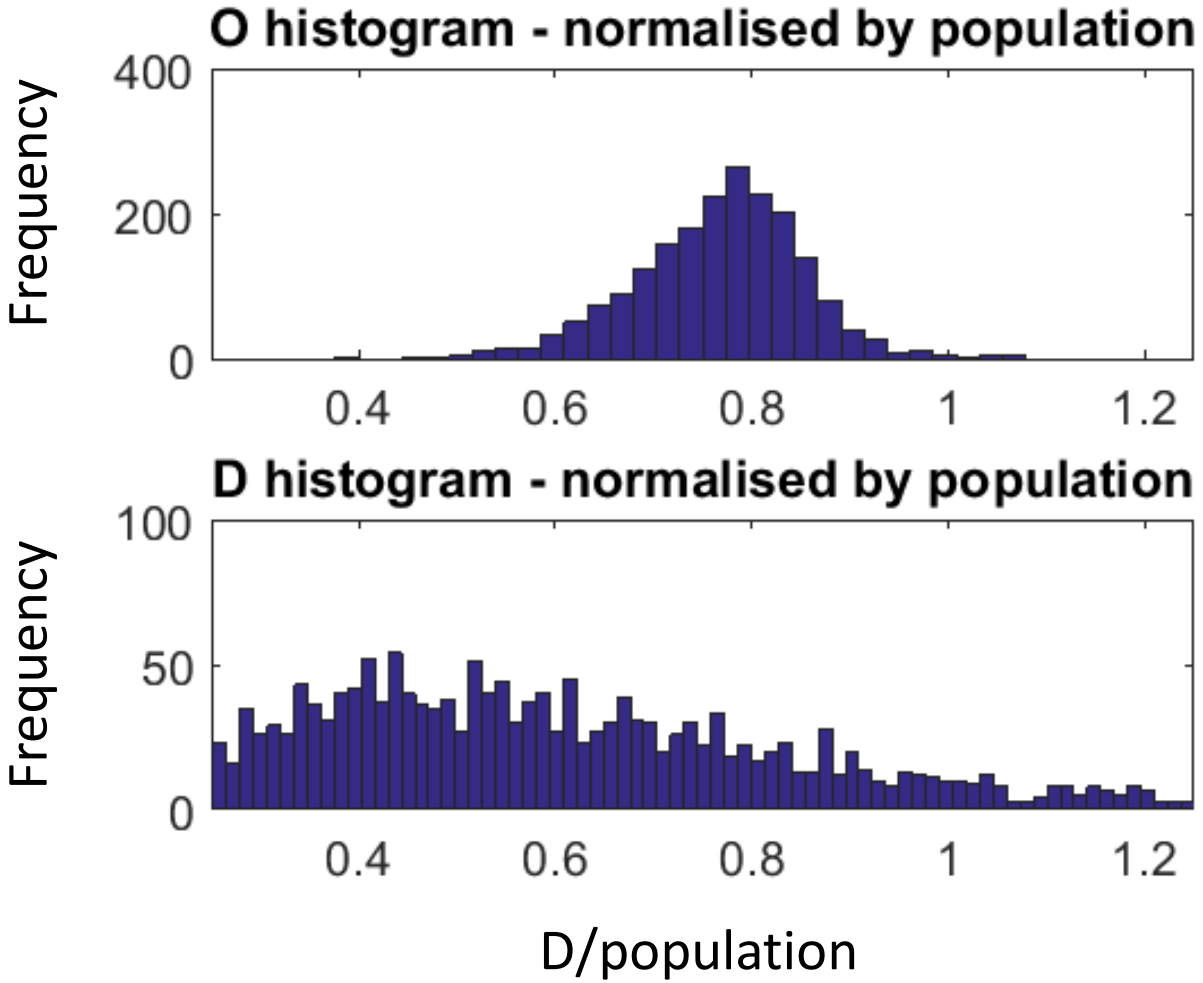
O and D histograms – normalised by pop.

- Let us now revisit the normalized O and D histograms with our merged zones.
- N = 4000 zones



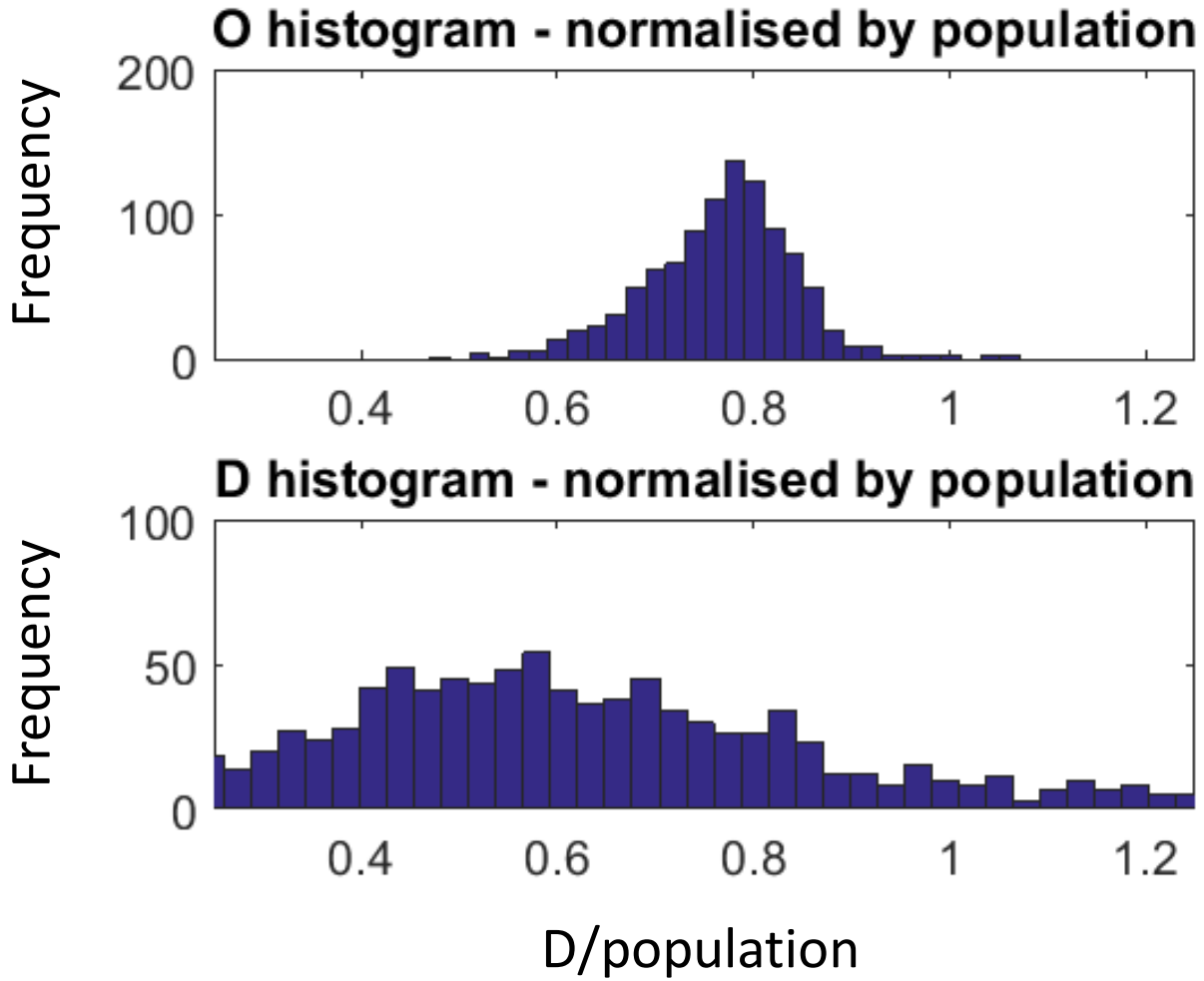
O and D histograms – normalised by pop.

- Let us now revisit the normalized O and D histograms with our merged zones.
- N = 2000 zones



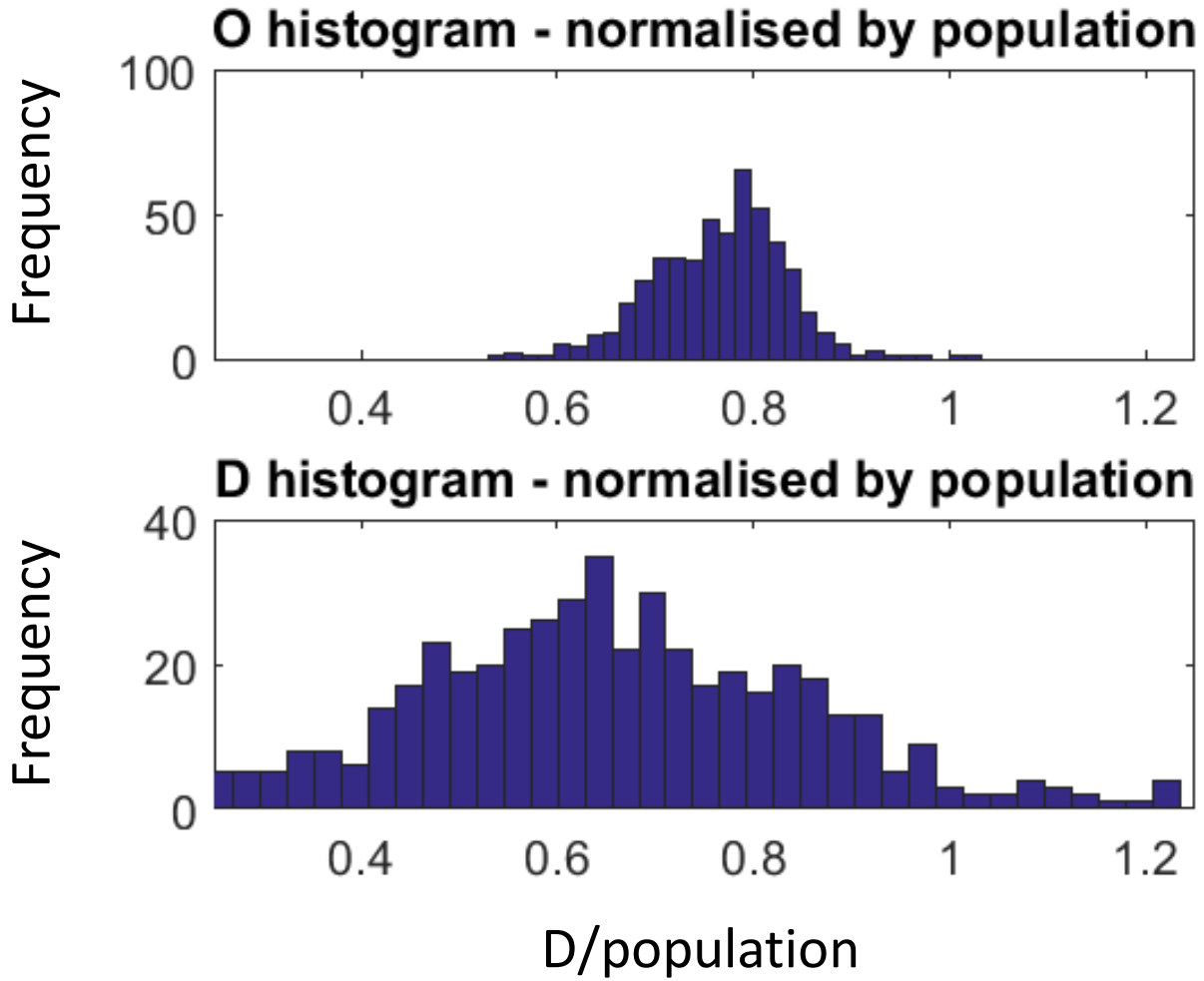
O and D histograms – normalised by pop.

- Let us now revisit the normalized O and D histograms with our merged zones.
- N = 1000 zones



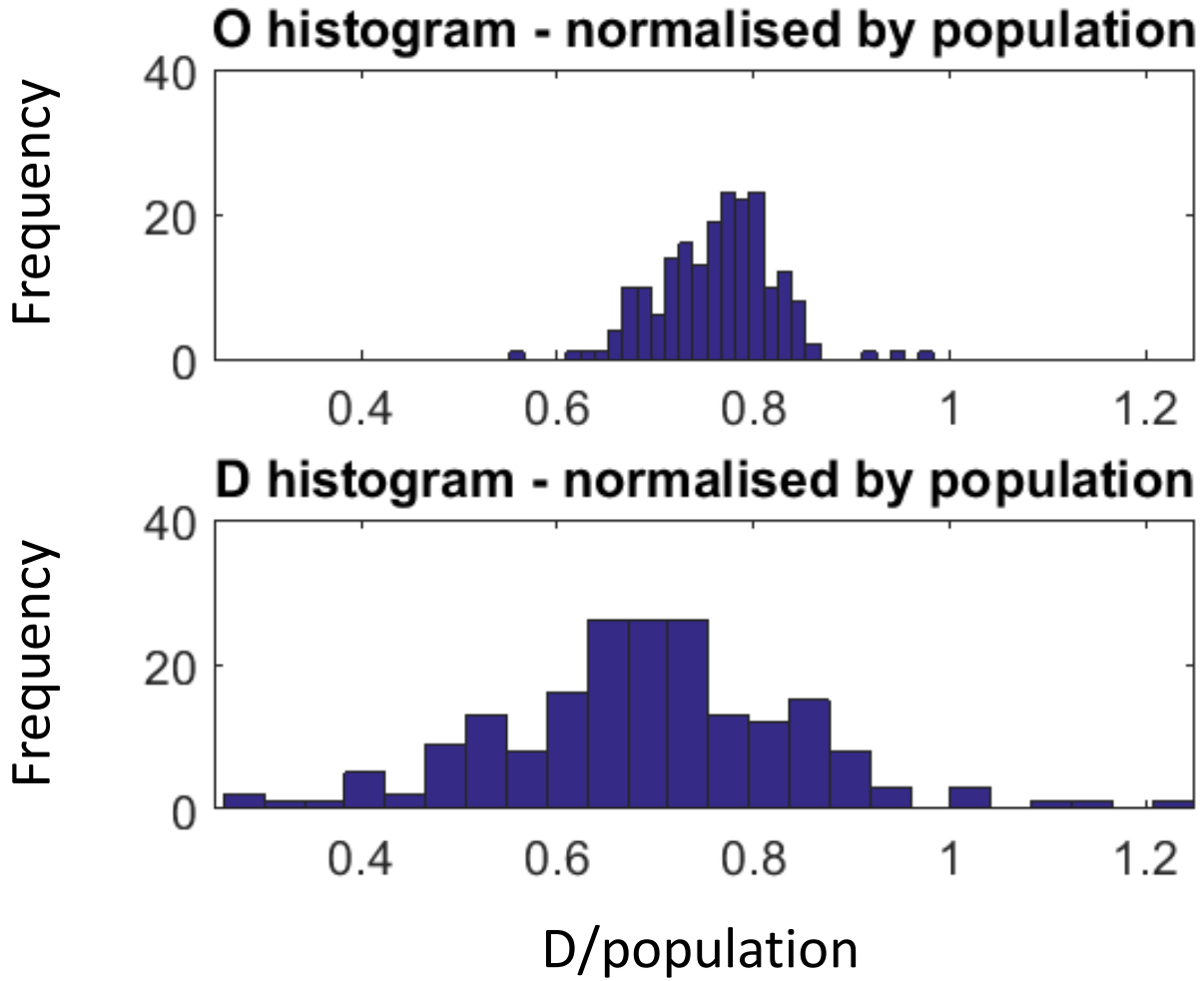
O and D histograms – normalised by pop.

- Let us now revisit the normalized O and D histograms with our merged zones.
- N = 500 zones



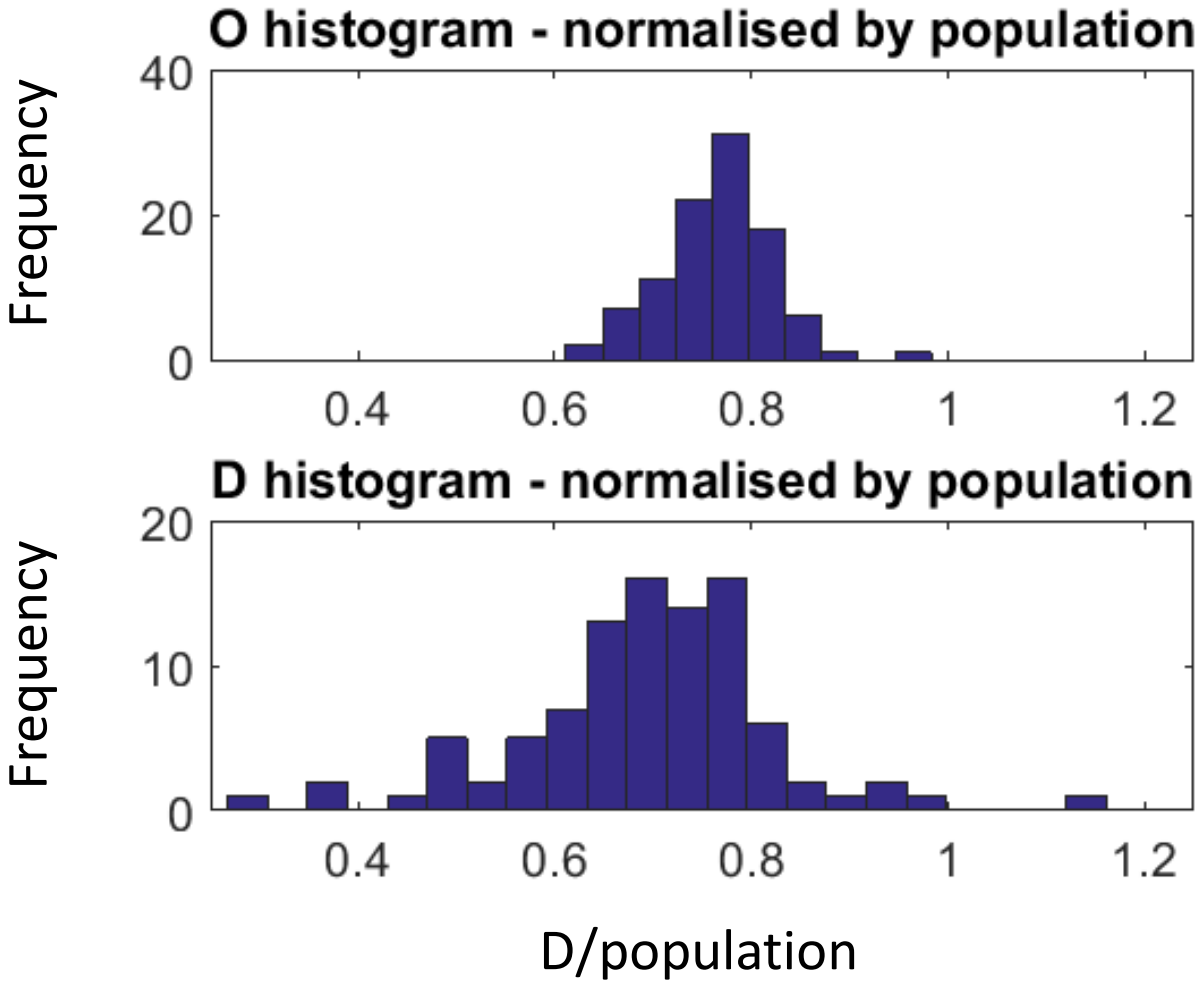
O and D histograms – normalised by pop.

- Let us now revisit the normalized O and D histograms with our merged zones.
- N = 200 zones



O and D histograms – normalised by pop.

- Let us now revisit the normalized O and D histograms with our merged zones.
- N = 100 zones
- Now our O and D histograms look very similar. How did that happen?



5. Social media data

Social media data:
Twitter geo-coded UK data set

Is the population of Twitter users representative to the UK as a whole?

UK Twitter population

- 18.6 million users (but only a fraction uses geo location)
- 59% male, 41% female.
- Majority of Twitter users in Great Britain are under 34 years of age.
- Twitter users tend to have a higher income and higher level of education compared to the population as a whole.

https://blog.hootsuite.com/twitter-demographics/

Census vs Twitter

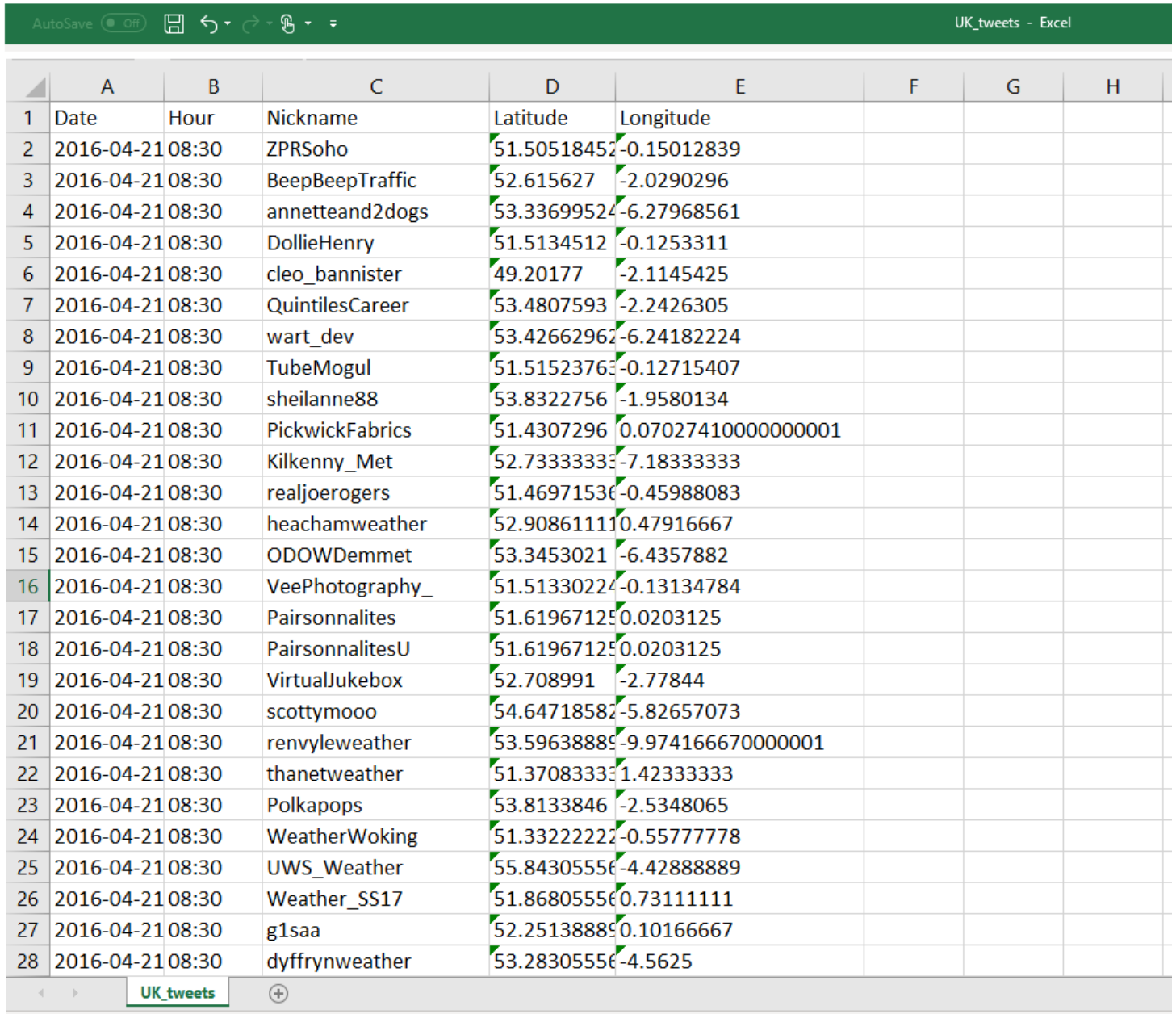
Main differences between the two data sets:

UK Census	**Twitter**
100% sample	Much smaller sample size
Unbiased	Strongly biased
Snapshot every 10 years	Live stream of data
Controlled data	Uncontrolled data

https://blog.hootsuite.com/twitter-demographics/

Twitter data set

- 170,000 geo-located tweets over a week's time (14-21 April 2016)
- Covering the UK plus Ireland (+a bit of France!)



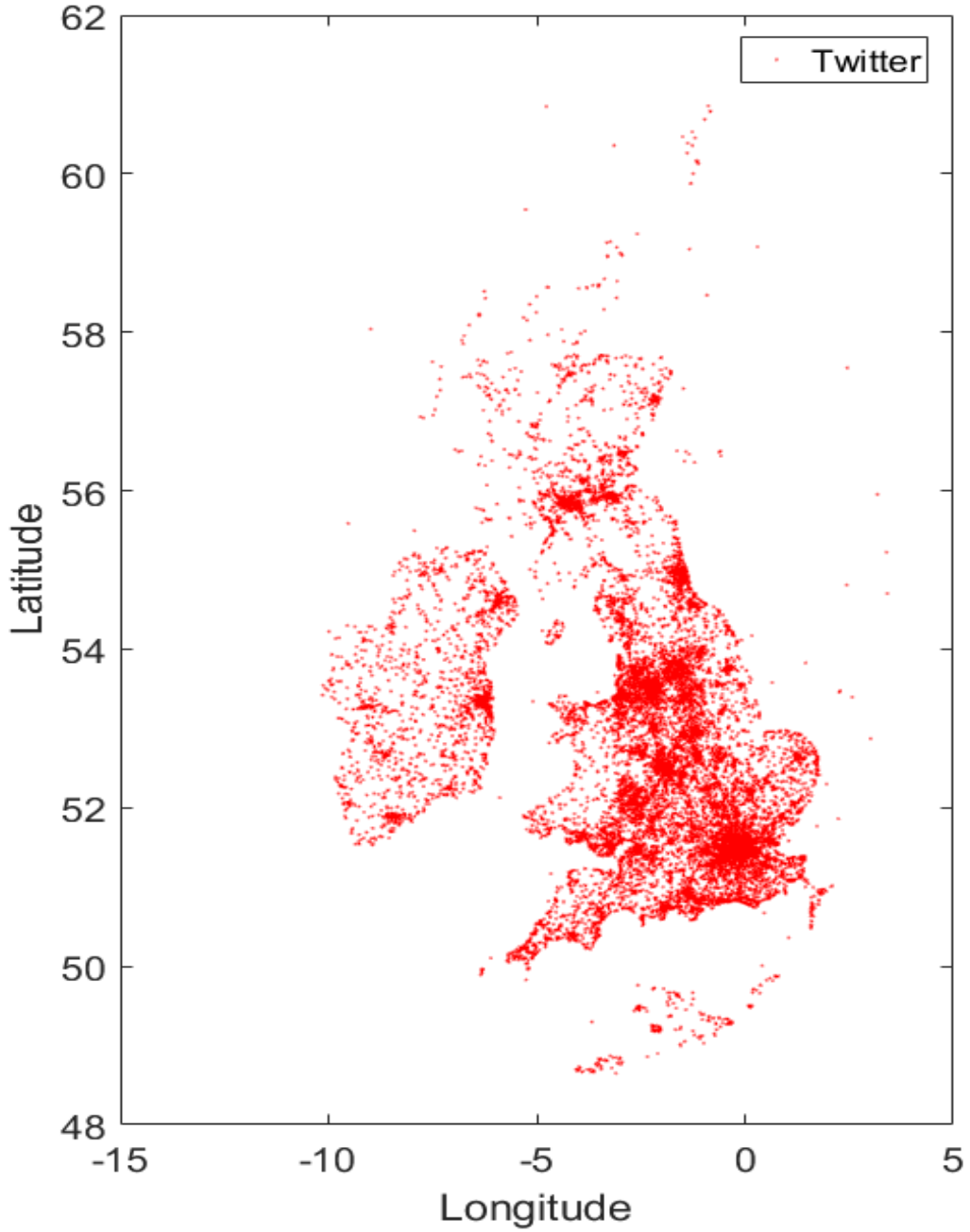
	A	B	C	D	E	F	G	H
1	Date	Hour	Nickname	Latitude	Longitude			
2	2016-04-21	08:30	ZPRSoho	51.50518452	-0.15012839			
3	2016-04-21	08:30	BeepBeepTraffic	52.615627	-2.0290296			
4	2016-04-21	08:30	annetteand2dogs	53.33699524	-6.27968561			
5	2016-04-21	08:30	DollieHenry	51.5134512	-0.1253311			
6	2016-04-21	08:30	cleo_bannister	49.20177	-2.1145425			
7	2016-04-21	08:30	QuintilesCareer	53.4807593	-2.2426305			
8	2016-04-21	08:30	wart_dev	53.42662962	-6.24182224			
9	2016-04-21	08:30	TubeMogul	51.51523763	-0.12715407			
10	2016-04-21	08:30	sheilanne88	53.8322756	-1.9580134			
11	2016-04-21	08:30	PickwickFabrics	51.4307296	0.07027410000000001			
12	2016-04-21	08:30	Kilkenny_Met	52.73333333	-7.18333333			
13	2016-04-21	08:30	realjoerogers	51.46971536	-0.45988083			
14	2016-04-21	08:30	heachamweather	52.90861111	0.47916667			
15	2016-04-21	08:30	ODOWDemmet	53.3453021	-6.4357882			
16	2016-04-21	08:30	VeePhotography_	51.51330224	-0.13134784			
17	2016-04-21	08:30	Pairsonnalites	51.61967125	0.0203125			
18	2016-04-21	08:30	PairsonnalitesU	51.61967125	0.0203125			
19	2016-04-21	08:30	VirtualJukebox	52.708991	-2.77844			
20	2016-04-21	08:30	scottymooo	54.64718582	-5.82657073			
21	2016-04-21	08:30	renvyleweather	53.59638889	-9.974166670000001			
22	2016-04-21	08:30	thanetweather	51.37083333	1.42333333			
23	2016-04-21	08:30	Polkapops	53.8133846	-2.5348065			
24	2016-04-21	08:30	WeatherWoking	51.33222222	-0.55777778			
25	2016-04-21	08:30	UWS_Weather	55.84305556	-4.42888889			
26	2016-04-21	08:30	Weather_SS17	51.86805556	0.73111111			
27	2016-04-21	08:30	g1saa	52.25138889	0.10166667			
28	2016-04-21	08:30	dyffrynweather	53.28305556	-4.5625			

UK_tweets

http://followthehashtag.com/datasets/170000-uk-geolocated-tweets-free-twitter-dataset/

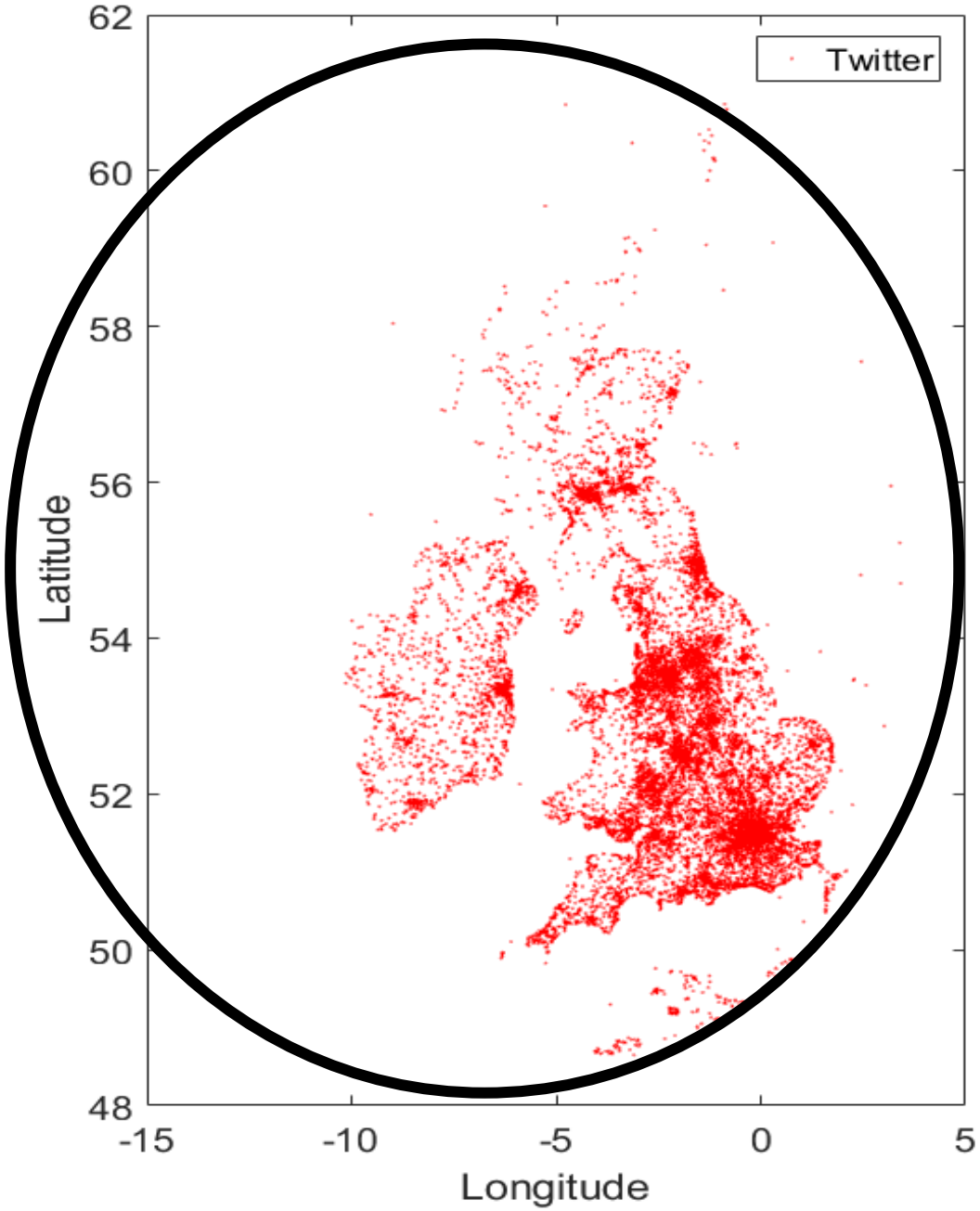
UK Twitter dataset

- By plotting the geo-location (longitude/latitude) of each tweet, it is clear that the **data contains a bit larger land mass than England and Wales only**



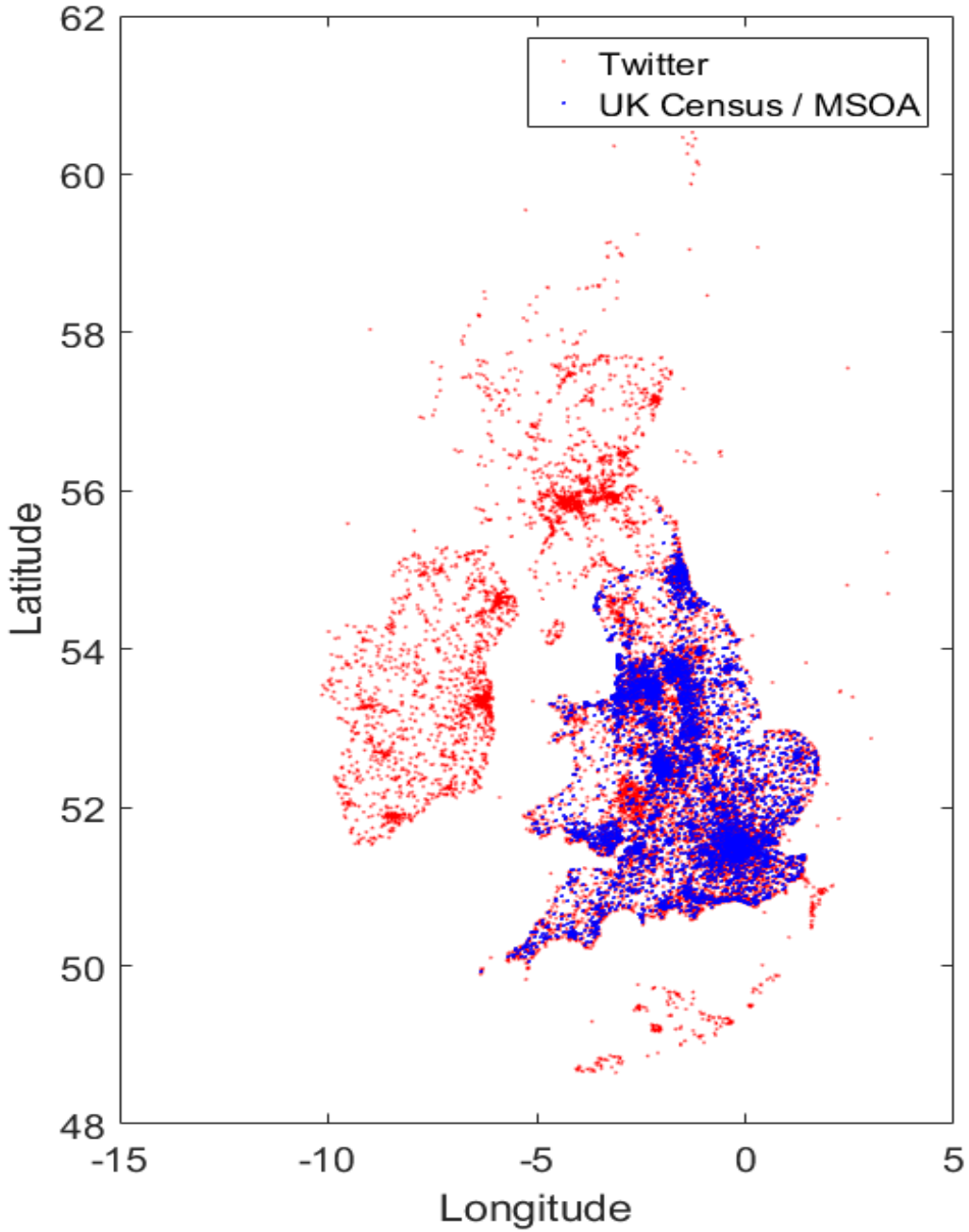
UK Twitter dataset

- By plotting the geo-location (longitude/latitude) of each tweet, it is clear that the data contains a bit larger land mass than England and Wales only
- **This is an artefact of the Twitter API which allows you to extract tweets within a radius of a central point.**



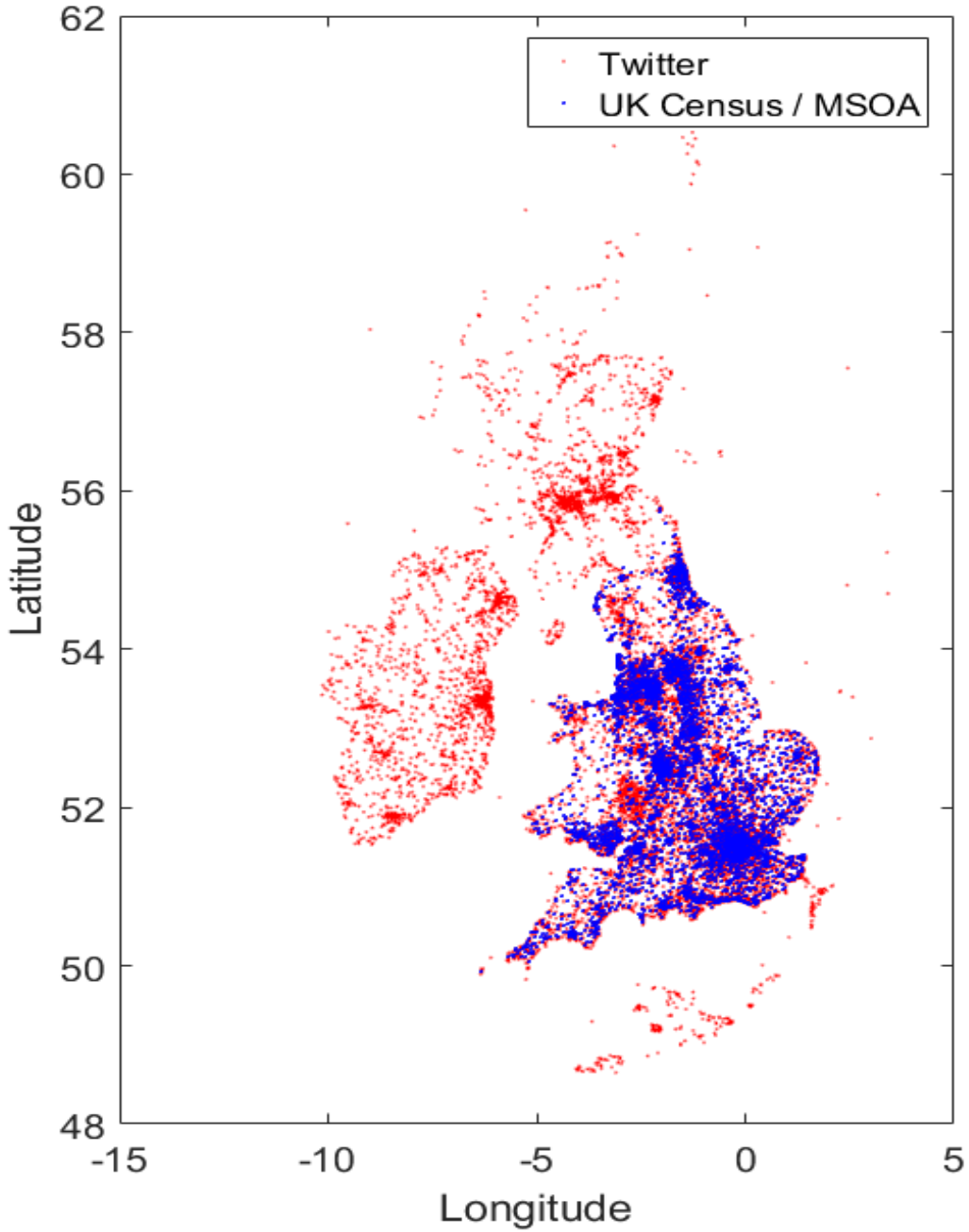
UK Twitter dataset

- Twitter data overlaid onto the UK Census / MSOA data.
- If we want to be able to compare the Twitter data with the UK Census data we will need to filter out all Tweets that fall outside of any of the England and Wales MSOA zone.



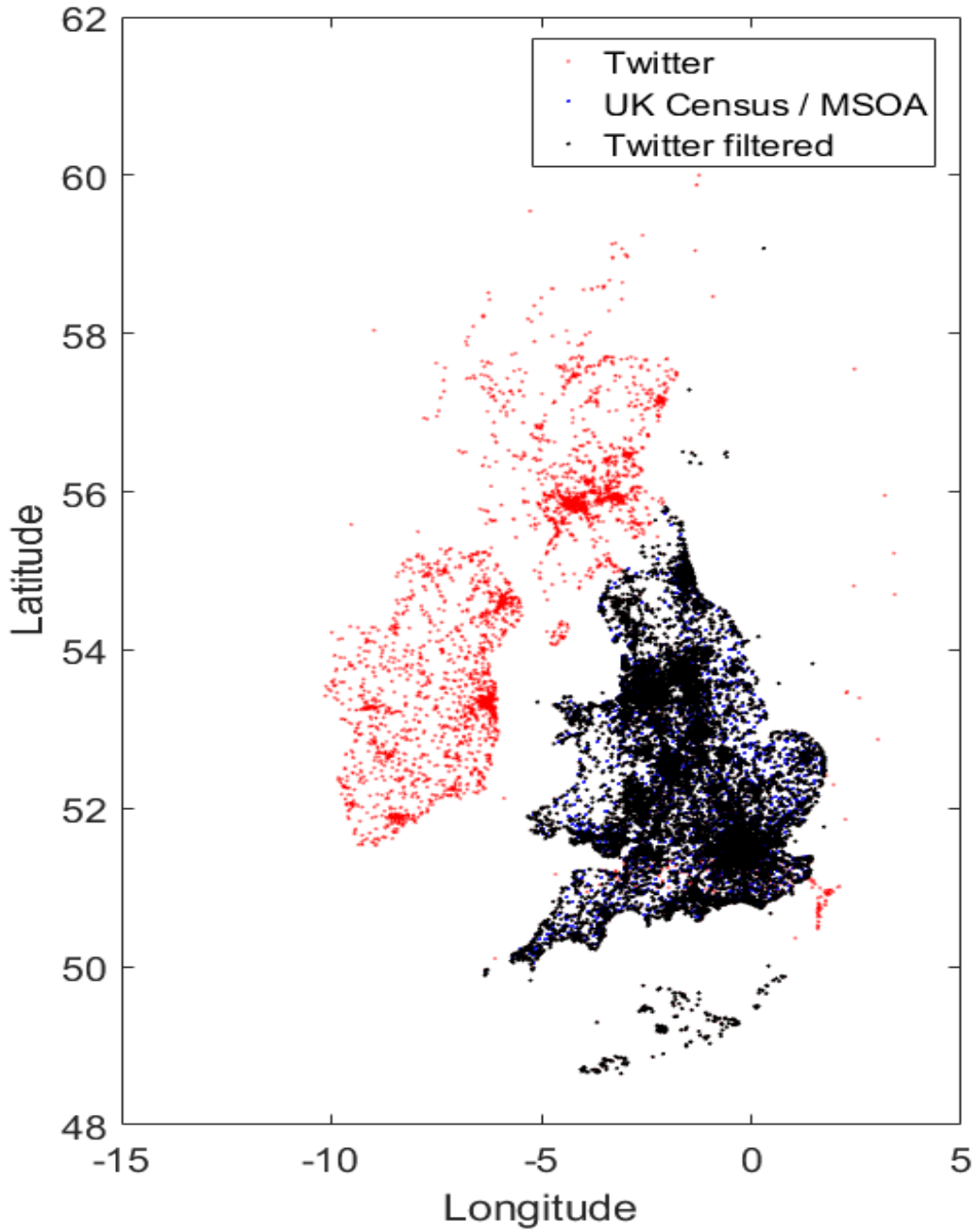
UK Twitter dataset

- If we had the complete **boundaries / shapefile data** for all MSOA zones it would have been trivial to see which zone each tweet fell into.
- However, on Tuesday we opted *not* to use the full boundary / shapefile dataset, so we need to be a bit creative now…



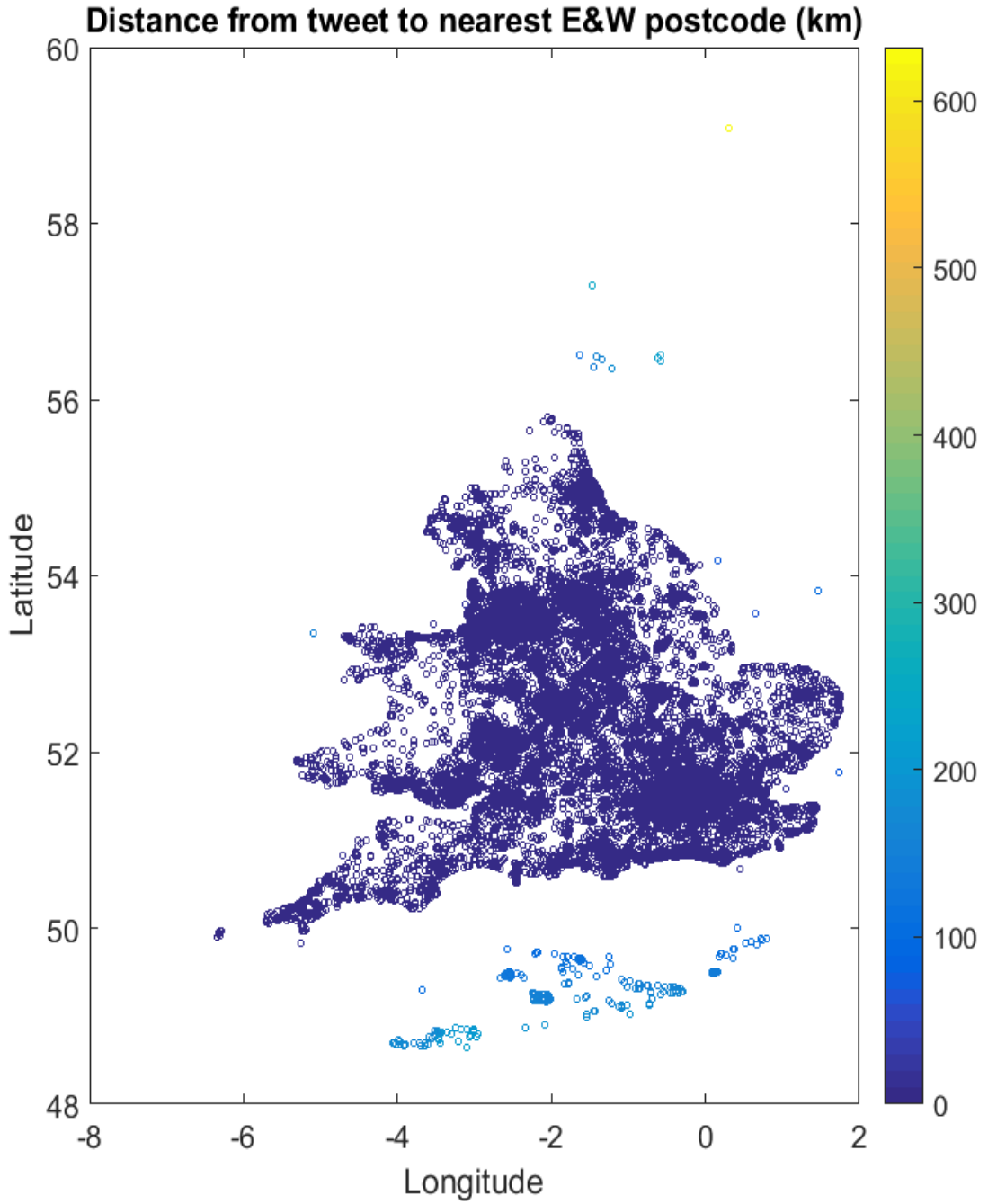
UK Twitter dataset

- Tweets from non-England and Wales postcodes are removed.



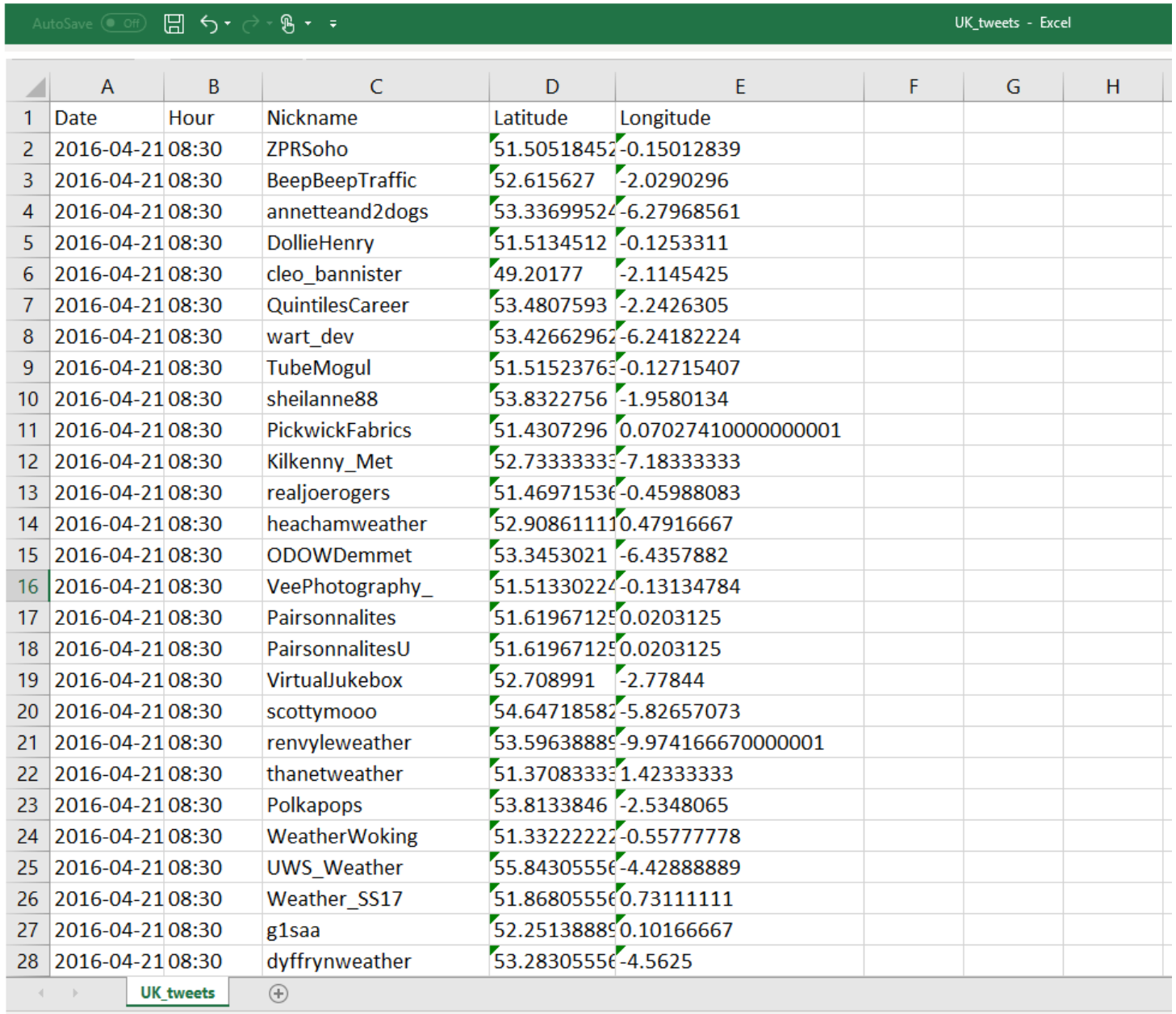
UK Twitter dataset

- The remaining few tweets (from France and ships?) can be filtered out if they fall too far away from an England or Wales post code.



Twitter data set

- How do we turn this data set this into something useful to us? (i.e. something pertaining to O_i and D_j)



AutoSave Off — UK_tweets - Excel

	A	B	C	D	E	F	G	H
1	Date	Hour	Nickname	Latitude	Longitude			
2	2016-04-21	08:30	ZPRSoho	51.50518452	-0.15012839			
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UK_tweets

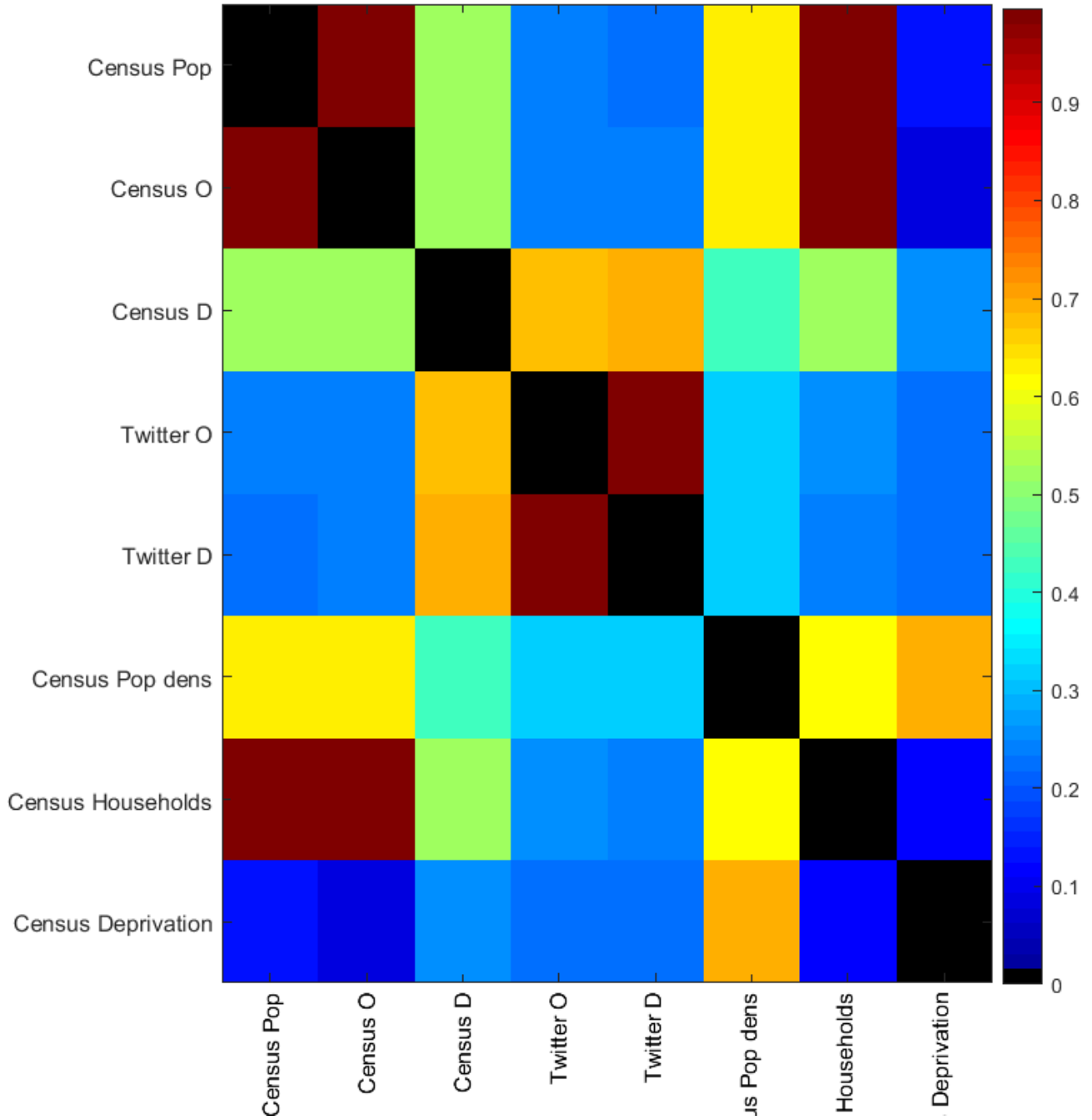
Transforming the Twitter dataset

In order to make the Twitter dataset useful to us, we need to make a few assumptions and then do something like this:

- **Option 1**: Assume that a tweet between 9am – 5pm on a weekday originates from a workplace, and a tweet outside of those hours is a tweet from a residential location.
- **Option 2**: Keep track of whenever the same unique user tweets in a new location (a new zone j), and then interpret that as a trip from the zone of the previous tweet *i* to the zone of the new tweet *j*.
- **Option 3**: …?

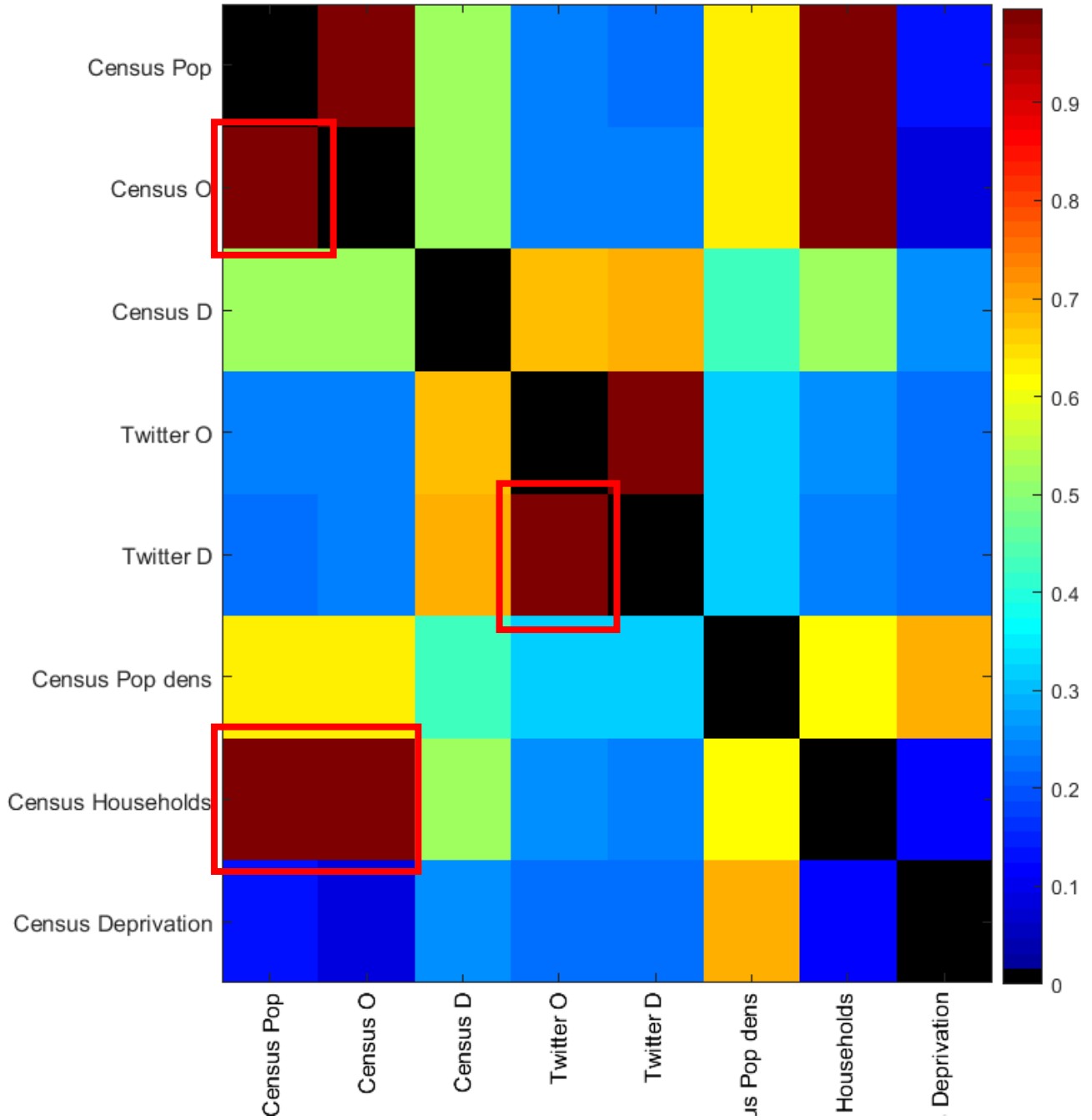
Correlation matrix

For N=2000 zones



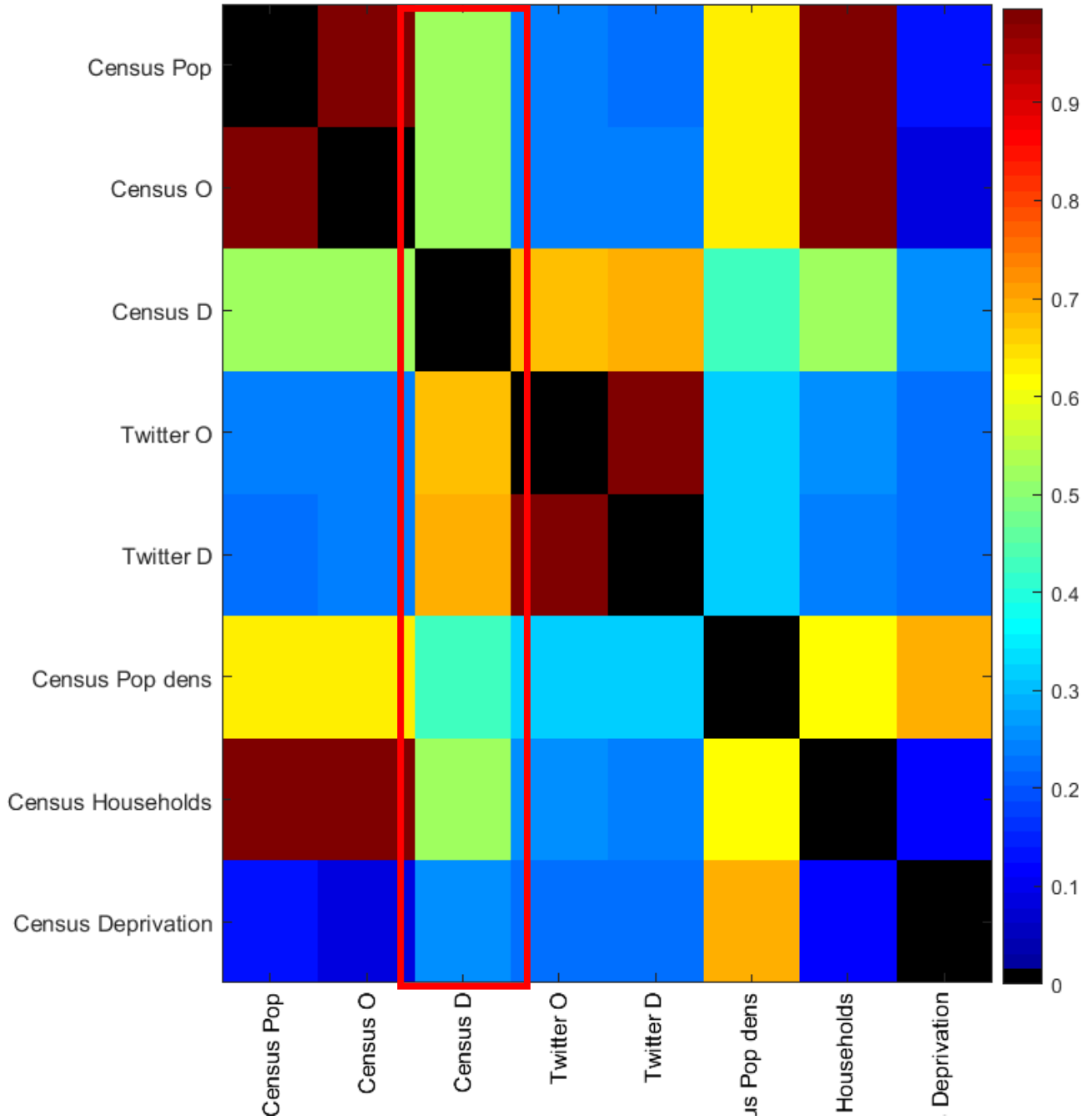
Correlation matrix

For N=2000 zones



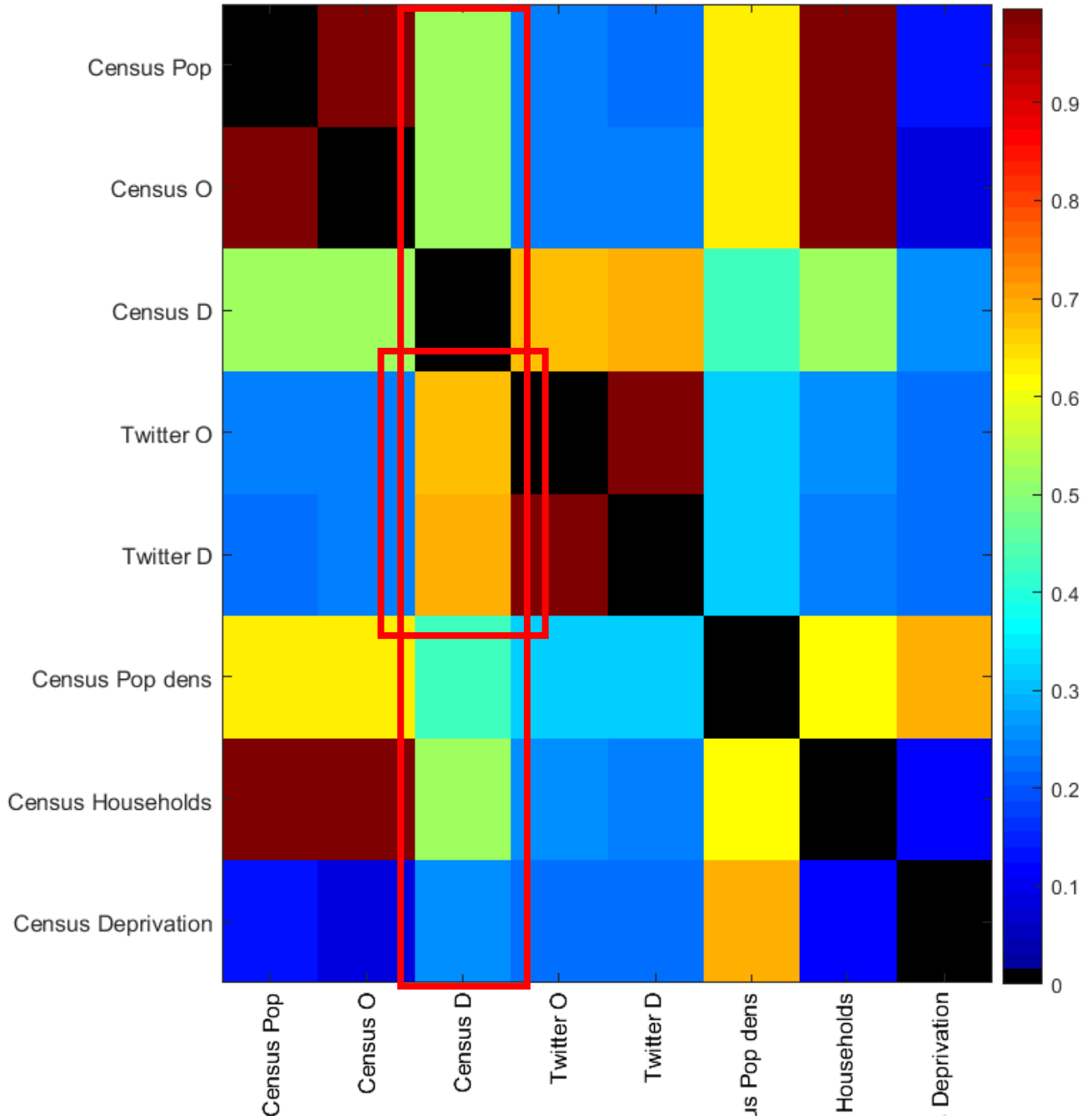
Correlation matrix

For N=2000 zones



Correlation matrix

For N=2000 zones



Population and Twitter D vs Census D

